

Economic Forecasting with an Agent-Based Model of the United States: Model, Calibration, and Out-of-Sample Performance against Statistical and DSGE Benchmarks

MarketsReplica research pipeline

17 August 2026

Abstract

We build an agent-based macroeconomic model of the United States on the framework of Poledna et al. [2023] and evaluate it under that paper’s own out-of-sample protocol: 61 quarterly forecast origins from 2010Q2 to 2025Q2, horizons of 1 to 12 quarters, 500 free-running stochastic paths per origin, and a twenty-column comparator field scored on identical cells. The model is calibrated from public data under a strict no-fitting rule — no behavioural parameter ever sees a forecast error — and completes the reference framework with three structural components that a United States implementation turns out to require: an opening supply–use system reconciled by biproportional (RAS) adjustment, because the published BEA tables do not balance once bridged across valuation bases and projected onto commodities; growth expectations formed by a random walk with drift, because the reference log-level autoregression is biased low by construction on trending data; and observed trend growth in supply — labour productivity, wages, and capacity efficiency advancing at an origin-local estimate of measured productivity-plus-labour-force growth — without which a fixed-population model must either exhaust its labour pool or drain its capacity headroom. The comparator field contains ten statistical benchmarks and two dynamic stochastic general equilibrium (DSGE) models re-estimated recursively at every origin, including a full Smets and Wouters [2007] system granted more data than any other column. On the headline pair {real GDP growth, GDP-deflator inflation} the ABM attains the best weighted RMSE ratio of all twenty columns in all three sample tracks (0.828, 0.828, 0.792 against a VAR(1) anchor); Romano–Wolf multiplicity-controlled inference separates it from both DSGE columns at the 5 % level in every track, while its margin over the best statistical columns remains marginal (adjusted $p \approx 0.08$). Its predictive densities are the sharpest in the field by continuous ranked probability score, but its GDP-deflator intervals are under-dispersed (90 % coverage 0.70–0.79), so a preregistered superiority bar fails on that single interval-calibration leg and the claim retained is the conditional one. The headline conclusion is unchanged under 2017 and 2012 input–output structures (weighted ratios move by at most 0.0007), and matched-seed component experiments attribute it: the accounting reconciliation carries most of the accuracy, the expectation specification most of the rest, and the supply-growth component — which repairs unemployment dynamics and long-run capacity validity — is deliberately headline-neutral. All of this is a revised-data, mixed-vintage diagnostic, not a real-time forecasting claim.

Keywords: agent-based macroeconomics; out-of-sample forecast evaluation; DSGE benchmarking; input–output reconciliation; balanced growth; forecast-comparison inference.

Reader’s note. This paper is written for a technically trained reader who is not an economist by training. Economic machinery is defined at first use, and the accounting identities that carry the argument are written out rather than assumed. Every number is traceable to a committed artifact; Appendix A gives the paths, hashes and commands. The repository’s development reports additionally

record how the model reached its present form; this paper presents the model as it stands and the evidence for each component, not the history.

1 Introduction

1.1 The Poledna programme

Macroeconomic forecasting is dominated by two model classes. *Time-series* models — vector autoregressions (VARs) and their Bayesian variants — treat the economy as a low-dimensional stochastic process and extrapolate its estimated dynamics; they are accurate at short horizons and structurally mute. *Dynamic stochastic general equilibrium* (DSGE) models [Smets and Wouters, 2007] write down a small number of optimising representative agents, impose market clearing at every date, and estimate the resulting cross-equation restrictions; they are structurally interpretable and, at forecast horizons, rarely better than a well-specified VAR.

Agent-based models (ABMs) take a third route [Farmer and Foley, 2009]. They simulate a population of heterogeneous entities — firms, households, a bank, a government — each following explicit behavioural rules, and let aggregates emerge from decentralised interaction. Markets do not clear by assumption; they clear, or fail to clear, through search and matching, with rationing as an observable outcome. For two decades this class was excluded from the forecasting conversation on the grounds that it was unfalsifiable in the only way that matters: nobody had run one against the standard benchmarks on a rolling out-of-sample grid.

Poledna et al. [2023] closed that gap. They calibrated a large ABM of Austria one-to-one against the national accounts, ran free-running forecasts from quarterly origins over 2010Q2–2016Q4 at horizons $h \in \{1, 2, 4, 8, 12\}$ with 500 stochastic paths each, and scored the ensemble mean against AR, VAR and Bayesian DSGE benchmarks. The ABM was competitive, and better than the DSGE at most horizons. That result is the reference point of the present work, and its protocol is the one reproduced here.

1.2 What this paper does

This paper ports that programme to the United States and answers its question at full strength: *does the ABM forecast the U.S. economy better than statistical and equilibrium challengers, on identical cells, under a protocol frozen in advance?* Five things are new relative to a naive port, and they organise the paper.

1. **The model, completed for the United States** (§2). The BeforeIT implementation [Glielmo et al., 2025] of the Poledna framework is retained whole; three structural components are added, each forced by a property of the U.S. setting and each calibrated from observations or accounting identities alone: a reconciled opening supply–use system, drift-form growth expectations, and observed trend growth in supply — productivity, wages and capacity efficiency advancing at a rate estimated from measured productivity and labour-force growth, the latter carrying the demographics-and-immigration margin.
2. **Calibration under a no-fitting firewall** (§3–§4). Every parameter is read from public data, computed from an identity, or estimated from pre-origin history by a fixed rule; no behavioural parameter ever sees a forecast error, and the claim is verified numerically rather than asserted.
3. **A twenty-column comparator field** (§6). Ten statistical benchmarks; two DSGE models re-estimated by posterior mode at every origin — an An and Schorfheide [2007]-type small New Keynesian model and a full 49-state Smets and Wouters [2007] system that is deliberately granted more history and more series than any other column; and the ABM’s mean and median ensemble columns. All columns are scored on identical common cells over 61 origins and three sample tracks.
4. **Results with formal inference and densities** (§7). Point standings with HLN-corrected Diebold–Mariano tests, Romano–Wolf multiplicity control and a model confidence set; exact

ensemble CRPS and interval coverage from the retained predictive draws; unemployment scored and reported honestly; and a structural-vintage robustness check that re-runs the whole exercise under 2017 and 2012 input–output structures.

5. **Component attribution** (§8). Matched-seed, identical-cell experiments that switch each component off — and two demand-side investment designs that were built and rejected on stability grounds — so that the headline accuracy is attributed to mechanisms rather than asserted of the model as a whole.

1.3 What the reader should take away, and what they should not

The ABM ranks first of twenty on the headline target pair in all three sample tracks, its margin over both DSGE columns is significant at the 5 % level under multiplicity control, and its predictive densities are the sharpest in the field by CRPS. That is a real result on a matched grid, and it is also a *revised-data, mixed-vintage diagnostic*: the data are current-vintage throughout, the annual structure is carried backwards to historical origins (a dependence measured directly in §7.8 and found immaterial for the input–output block), and no real-time information set was reconstructed. The margin over the best *statistical* columns is marginal under multiplicity control (adjusted $p \approx 0.08$); the GDP-deflator’s predictive intervals are under-dispersed, which fails one leg of the preregistered superiority bar; and on the unemployment *level* the ABM loses to simple persistence models, for a measured initialisation reason. These limits are stated in §10 and are not incidental to the result; they are part of it.

2 The model

This section is a compact overview sufficient to read the results. The full equation system is in Poledna et al. [2023] and its implementation in Glielmo et al. [2025]; the behavioural block is inherited whole, plus two flag-gated specification choices (the growth expectation of §2.3 and the trend-growth component of §2.4) that are off by default, so the Austrian and Italian calibrations of the library remain bit-identical.

2.1 Agents and the flow of a quarter

The economy contains six classes of agent.

Firms.

$\sum_s I_s$ firms distributed over S industries. Each firm i in industry s produces a single commodity with a *Leontief* technology [Leontief, 1936]: output requires labour, intermediate inputs and capital services in fixed proportions, with no substitution between them. Concretely, output is capped by the binding factor,

$$Y_i = \min \left(\underbrace{\alpha_s N_i}_{\text{labour}}, \underbrace{\beta_s M_i}_{\text{materials}}, \underbrace{\kappa_s K_i}_{\text{capital}} \right), \quad (1)$$

where N_i is employment, M_i the intermediate-input bundle, K_i the capital stock, and $(\alpha_s, \beta_s, \kappa_s)$ are industry productivities read off the input–output table. The third term is the *capacity envelope*; how it is initialised, and how it grows, are two of the design decisions this paper measures (§3.2, §2.4).

Households.

Worker households supply labour and consume; inactive households receive transfers and consume; firm owners receive dividends. Consumption follows a propensity rule on disposable income and liquid wealth, allocated across commodities by a fixed basket b_g^{HH} .

Table 1: The five model relations the results turn on. Notation follows Poledna et al. [2023]; δ_s is the depreciation rate, ω the assumed normal capacity utilisation, Q_i^d desired sales, Q_i^s target production, and g the calibrated quarterly trend-growth rate of §2.4 ($g = 0$ reproduces the reference model exactly).

Relation	Form and role
Production, Leontief with a capacity cap	$Y_i = \min(\alpha_s N_i, \beta_s M_i, \kappa_s K_i)$. The min is <i>one-sided</i> : a firm below its cap cannot lend spare capacity to a firm above it.
Target output	$Q_i^s = Q_i^d (1 + \gamma_e)$, where γ_e is the economy-wide expected growth rate of §2.3.
Desired investment	$I_i^d = (\delta_i / \kappa_i) \min(Q_i^s, K_i \kappa_i)$, and the law of motion $K' = K - (\delta / \kappa) Y + I$. When firms obtain their desired investment and produce at target these give $K' = K$ exactly: investment is <i>replacement-only</i> , and book capital is a constant.
Opening capacity	$\kappa_s = \tau$ (industry output) / (fixed assets) / ω and $K_i = Y_i / (\omega \kappa_i)$, so $K_i \kappa_i = Y_i / \omega$ identically. With $\omega = 0.85$ every firm opens at 85 % utilisation with $1/0.85 - 1 = 17.6$ % headroom.
Trend supply growth	$\alpha_s, \bar{w}_s, \kappa_s, \delta_s$ each grow by the factor $(1 + g)$ at the start of every simulated quarter (§2.4). Unit labour cost $\bar{w}/\alpha$ and the replacement share δ/κ are invariant by construction; capacity $K\kappa$ grows at g on constant book capital.

A bank.

One consolidated commercial bank intermediates deposits and loans subject to a capital-adequacy constraint, and rations credit when the constraint binds.

A central bank.

Sets the policy rate by a Taylor rule — a feedback rule raising the rate when inflation or an activity measure exceeds target — with inertia ρ , an intercept r^* , and response coefficients ξ_π and ξ_γ .

A government.

Collects taxes, pays wages, transfers and unemployment benefits, consumes a commodity basket c_g^G , and issues debt.

A rest-of-world block.

Exports absorb a basket c_g^E ; imports supply c_g^I . In this calibration the block is exogenous and is driven by estimated univariate processes.

Within a quarter, firms form expectations, choose target output and desired inputs, post vacancies and demand credit; the labour, credit and goods markets then clear through decentralised *search and matching* — buyers meet a random subset of sellers, transact at posted prices, and any unmet demand is simply not satisfied. Rationing is therefore a first-class observable of the model, reported here as *fill rates*. Insolvent firms exit and are replaced.

2.2 The behavioural core in five relations

Table 1 lists the relations that carry the argument of this paper. The remaining several dozen — price and wage setting, credit allocation, the bank’s balance sheet, the fiscal block — are unchanged from the reference implementation and are not reproduced.

Two structural facts about this core shape everything downstream, and both are measured rather than assumed in §8. First, the capacity envelope is anchored to the *opening output composition*: if the opening demand system asks sectors for quantities inconsistent with that composition, the one-sided min prevents idle capacity in one sector from compensating a binding ceiling in another, and nothing in the behavioural block can reconcile the two. The opening

Table 2: Why the reference growth-expectation estimator is replaced: the log-level AR(1) at six origins. T' is the length of the model’s own output history at the origin; $\hat{\alpha}$ the fitted coefficient; the last column the fraction of the in-sample trend the expectation delivers. Pinning $\hat{\alpha} = 1$ — the drift form used here, on the same past-only data — recovers +2.33, +2.39, +2.36, +2.42, +2.37 and +2.40%: the trend, at every origin.

Origin	T'	$\hat{\alpha}$	γ_e (%)	trend (%)	γ_e/trend
2012Q4	64	0.9713	+0.745	+2.331	0.32
2014Q4	72	0.9791	+0.914	+2.385	0.38
2016Q4	80	0.9834	+1.035	+2.359	0.44
2019Q4	92	0.9906	+1.469	+2.415	0.61
2022Q4	104	0.9900	+1.240	+2.366	0.52
2024Q4	112	0.9930	+1.523	+2.401	0.63

accounts must therefore balance — an accounting precondition the calibration of §3.2 enforces exactly. Second, with replacement-only investment the envelope never moves on its own: over every experimental variant run for this work, $K_{\text{end}}/K_1 = 1.0000\text{--}1.0009$. Whatever trend growth the economy is to have must therefore enter through the productivity coefficients, which is what §2.4 does — and §8.4 reports why the alternative, letting investment itself respond to demand, was built and rejected.

2.3 Growth expectations: a random walk with drift

Firms extrapolate aggregate demand: target output is desired sales scaled by an economy-wide expected growth rate γ_e , re-estimated every quarter from the model’s own realised output history. The specification matters more than it looks. The reference implementation estimates an ordinary-least-squares AR(1) with constant on the *log level* of output — a series that is, in statistical terms, integrated with drift (its level wanders; its growth rate is roughly stationary around a positive mean). For a log-linear trend $x_t = a + gt$, that estimator has the closed-form expectation

$$\gamma_e \approx g \left[1 - (1 - \hat{\alpha}) \frac{T}{2} \right], \quad (2)$$

where $\hat{\alpha}$ is the fitted autoregressive coefficient and T the estimation window. Because $\hat{\alpha}$ sits below one in any finite sample — the Kendall [1954] small-sample bias, compounded by the level shocks of 2008–09 and 2020 — the bracket is strictly less than one: *the estimator delivers only a fraction of the trend it was shown*, and the fraction *falls* as the window lengthens. Measured across six representative origins the delivered fraction is 32–63% (Table 2), reproduced by Eq. (2) to 0.05 pp. Shorter windows do not help: they trade a smaller T against a more biased $\hat{\alpha}$.

The model therefore uses the correctly specified estimator for an integrated-with-drift series: a *random walk with drift*, $\hat{\alpha}$ pinned to one and the drift taken as average past log growth, on past data only. At the same six origins this recovers +2.33 to +2.42% annualised — the trend, at every origin. The choice consumes exactly one Normal variate per quarter on either branch, a deliberate property that keeps matched-seed component experiments aligned (§8); it is gated behind a model property that is default-off; and its measured contribution to forecast accuracy is reported in §8.1 rather than folded silently into the model.

2.4 Trend growth in supply

The reference model, ported literally, has no trend growth in supply anywhere: labour productivity α_s , the wage base $\bar{w}_s$, the labour force, and the capacity envelope $K\kappa$ are all frozen at their opening values for the entire simulation. For Austria over a twelve-quarter horizon this is a tolerable abstraction. For the United States it is not, and the failure is arithmetic, not aesthetic.

U.S. trend output growth of roughly 2 % per year has two measured supply-side sources — output per worker (productivity) and the number of workers (population and immigration) — in roughly comparable parts. A model in which neither source exists can grow only by working its fixed labour pool and fixed capacity harder: simulated demand growth must be met one-for-one by employment growth, so unemployment falls mechanically quarter after quarter until the pool is exhausted, while output growth drains the fixed 17.6 % capacity headroom until utilisation hits its ceiling. Both pathologies are measured directly in §8.3; each invalidates a different part of the model — the labour market within the forecast window, the growth mechanism beyond it.

The model therefore registers observed trend growth on the supply side, per origin, from data available at the origin. Let

$$g = \underbrace{\overline{\Delta \ln(\text{GDP}/\text{employment})}}_{\text{labour productivity}} + \underbrace{\overline{\Delta \ln(\text{labour force})}}_{\text{demographics and immigration}}, \quad (3)$$

each term a trailing 40-quarter mean ending at the forecast origin, computed from measured series (real GDP, civilian employment, civilian labour force; sources in §3.4). At the start of every simulated quarter the supply coefficients advance by this rate: α_s and $\bar{w}_s$ grow by $(1 + g)$ — so output per worker and the wage level track trend while *unit labour cost* $\bar{w}/\alpha$, the object that drives pricing, is invariant — and κ_s and δ_s grow by the same factor — so productive capacity $K\kappa$ tracks trend on constant book capital while the replacement share δ/κ (capital consumption per unit of output) is exactly stable. No demand flow is injected anywhere, and the accounting identity net $I = \Delta K$ holds trivially in book units.

Three properties make this the right closure rather than a convenient one. *It is observed, not fitted*: g is a trailing average of measured series, re-estimated at each origin by a fixed rule frozen before any forecast was scored; across the 61 origins it runs 1.5–2.3 % annualised, and its labour-force component — the immigration margin — tripled between the 2020 and 2025 origins (labour-force growth averaged 1.26 %/yr over 2022–2024 against 0.75 %/yr over 2011–2019), which the origin-local window picks up automatically. *It is balanced*: demand trends in the model (household income, government consumption, exports) already grow at observed rates through the estimated exogenous processes; giving supply the same measured trend removes a structural asymmetry rather than adding a stimulus. *It is falsifiable in components*: switching the mechanism off collapses unemployment and utilisation exactly as the arithmetic predicts, and the demand-side alternative — letting desired investment respond to the capacity gap — destabilises the economy; both experiments are reported in §8.

3 United States calibration and data

3.1 The calibration artifact

A *calibration artifact* is a single serialised object holding every parameter and initial condition the simulator needs: the technology matrix, final-demand baskets, agent counts, tax and benefit rates, opening stocks, and the quarterly exogenous series. It is built once from public data and then loaded read-only by the forecasting harness.

The classification comes from the BEA summary-level supply–use system [U.S. Bureau of Economic Analysis, 2025]. That source carries 68 observed commodities and 71 industries; the four retail industry columns (441, 445, 452, 4A0) are summed into a single model sector 4A0, giving 68 modelled industries matched one-to-one against the 68 commodities. Table 3 gives the resulting dimensions.

Table 3: United States calibration at scale = 10^{-5} . One modelled firm stands for 10^5 real ones; the reference Austrian exercise ran at 1:1.

Commodities G	68	Economically active households H^{act}	1 826
Industries S (from 71 BEA industries)	68	Inactive households H^{inact}	1 005
Firms $\sum_s I_s$	130	Total population represented	$\approx 2\,831$
Government entities J	33	Foreign consumers L	65
Runtime: 0.085 s per 12-quarter path after compilation, single-threaded.			

3.2 The opening accounts must balance, and are made to

An input–output system published by a statistical agency balances *in the form in which it is published*: every commodity’s supply equals its uses, in one valuation basis, on one set of axes. The U.S. system is published at purchasers’ prices over non-identical industry and commodity axes, while the model’s coefficients are defined at basic prices on a symmetric commodity basis (§4 details the bridge). After the required valuation bridge and industry-to-commodity projection — and after the decision to take commodity imports from the measured BEA T262 import matrix rather than solving them as a residual, which removes the last free slack variable — the balance is no longer guaranteed. In the raw U.S. artifact it fails badly: modelled uses differ from supply by -70.6% to $+47.6\%$ across the 68 commodities, with a cross-sector standard deviation of 17.9 percentage points.

An equilibrium or time-series model would never notice. This model must: the opening demand composition and the opening capacity envelope are *both* anchored to the same accounts (§2), so a sector asked for 47% more than its recorded output sits permanently against its capacity ceiling while a sector asked for less idles, and the one-sided production constraint lets the shortfalls accumulate without offset. Left in place, this single accounting defect costs about -2.75 percentage points of annualised real-growth forecast bias; the measured anatomy is in §8.2.

The calibration therefore reconciles the use side exactly, by *biproportional adjustment* — the RAS method of Stone [1961] and Bacharach [1970], equivalently the minimum-relative-entropy projection: find the use matrix closest to the observed one, in the information-theoretic sense, that satisfies the accounting margins. Formally, with $U^0 \in \mathbb{R}_{\geq 0}^{G \times J}$ the observed use matrix over $G = 68$ commodities and $J = 72$ users (68 industry intermediate columns plus consumption C , government G , investment I and exports X),

$$U^* = \arg \min_{U \geq 0} \sum_{g,j} U_{gj} \log \frac{U_{gj}}{U_{gj}^0} \quad \text{s.t.} \quad \sum_j U_{gj} = r_g \quad \forall g, \quad \sum_g U_{gj} = c_j \quad \forall j, \quad (4)$$

with row controls $r_g = \text{industry output}_g + \text{measured imports}_g$ and column controls c_j the measured budgets. The solution has the biproportional form $U_{gj}^* = \rho_g U_{gj}^0 s_j$, is obtained by alternately scaling rows and columns, and preserves structural zeros exactly. Crucially, the measured import levels are *retained*: the reconciliation works around the measured trade data rather than discarding it.

One accounting choice is required before RAS can run at all: row and column controls must sum to the same total, and in the raw artifact they do not — uses exceed supply by 1.0104% in aggregate. That excess decomposes *exactly* (to 5.6×10^{-9}) into two measurement facts of the source data: the accounts’ own production-versus-expenditure statistical discrepancy (\$442.9bn, 1.57% of GDP) and negative net non-dwelling fixed investment (capital consumption exceeding non-dwelling gross fixed capital formation by \$88.2bn). One control must move, and the calibration anchors on the production account, scaling the four final-demand aggregates

Table 4: The RAS reconciliation as executed, from the committed reconciliation report. The adjustment moves composition, not aggregates: every column budget is preserved to 10^{-13} after the λ scaling of Eq. (5).

Flow system	68 commodities $\times$ 72 users
Structural zeros preserved	1 549 / 4 896 cells (31.6 %)
Row controls	industry output + measured imports, sum 5.25699×10^7
Column controls	measured budgets, held exactly, sum 5.25699×10^7
Convergence	623 iterations; final max relative control error 9.996×10^{-14}
Balance after RAS	$\max \text{uses}_g / \text{supply}_g - 1 = 5.55 \times 10^{-16}$; cross-sector s.d. 1.96×10^{-16} (was 0.1790)
Balance after re-derivation	$\max \text{uses}_g / \text{supply}_g - 1 = 1.00 \times 10^{-13}$
Total use adjustment	$\sum d = 6.786 \times 10^6$ (12.78 % of total uses)
Cellwise adjustment	$\sum dU = 7.423 \times 10^6$ (13.98 % of total flow)
<i>Coefficient movement received by the simulator</i>	
Technology matrix a_{sg}	$\max \Delta a = 0.1408$; mean $ \Delta a = 0.00226$
Household basket b_g^{HH}	$\max \Delta \text{share} = 0.0158$; L_1 distance 0.0970
Government basket c_g^G	$\max \Delta \text{share} = 0.1450$; L_1 distance 0.3204
Investment basket b_g^{CF}	$\max \Delta \text{share} = 0.0401$; L_1 distance 0.1049
Export basket c_g^E	$\max \Delta \text{share} = 0.0309$; L_1 distance 0.1197
Import basket c_g^I	unchanged

(and the two components that set the investment budget) by

$$\lambda = \frac{\text{supply} - \text{intermediates}}{C + G + I + X} = 0.983460. \quad (5)$$

λ is fixed by the accounting identity alone; its semantics are recorded verbatim in every result manifest. The alternative closure — hold every measured budget and let a uniform 1.01 % excess-demand wedge survive — was built and measured too; it overshoots, because a uniform 1 % excess demand is itself a growth impulse (§8.3).

Figure 1 shows the before-and-after distribution. The adjustment is not uniform across users: as a share of its own budget, the government column moves most (31.0 %), then intermediates (15.3 %), exports (12.1 %), investment (10.0 %) and households (9.8 %). The largest single commodity adjustments are professional and technical services (uses cut 32.3 %, closing a +47.6 % gap), wholesale trade (raised 42.5 %, closing a -29.8 % gap) and state and local government (raised 26.2 %).

3.3 United States-specific choices in the artifact

Three further choices depart from the reference specification.

Explicit measured trade (`use_explicit_trade = true`).

Commodity imports are taken from the measured BEA T262 import matrix rather than being solved as the supply–use residual. This preserves the measured import level and is closer to the reference paper’s stated intent of reading trade off the input–output table; it is also what removes the residual degree of freedom and makes the reconciliation of §3.2 necessary. The reconciliation retains it.

Opening inventories are zero.

Sector inventories open at $S_i(0) = 0$, as in the reference specification. (An earlier artifact derived opening inventories from the positive part of the negated statistical discrepancy; a signed accounting residual is not an inventory stock, and once the balance clears there is no residual to promote.)

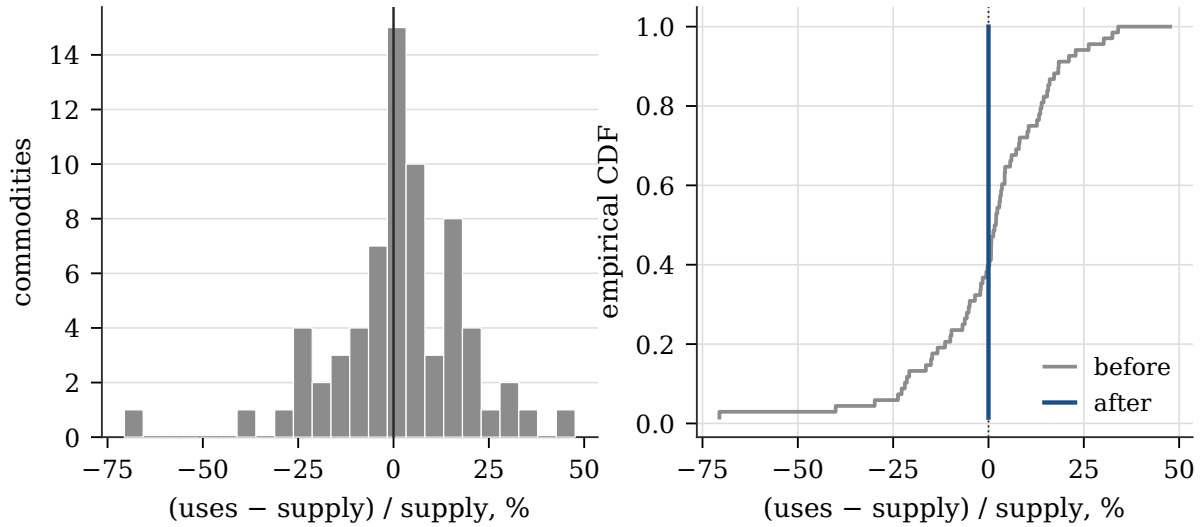


Figure 1: F5 — the opening commodity balance, before and after reconciliation. Left: the distribution of $(\text{uses}_g - \text{supply}_g) / \text{supply}_g$ over the 68 commodities of the raw artifact, spanning -70.6% to $+47.6\%$ with a standard deviation of 17.9 percentage points. Right: the empirical cumulative distribution before (grey) and after (blue). The post-reconciliation distribution is a step function at zero: every commodity clears to within 10^{-13} . Data: `reconciliation_by_commodity_rho1.csv`.

Policy rate on a quarterly basis.

The legacy field named `euribor` holds the U.S. policy rate, stored as $r = (\text{annual rate} + 1)^{1/4} - 1$. Comparison against the annualised-percentage-point effective federal funds rate (EFFR) therefore requires $100((1 + r)^4 - 1)$. At 2024Q4 this gives 4.571 % against a panel value of 4.65 %.

3.4 Trend-growth inputs

The trend rate g of Eq. (3) is computed from three measured series — real GDP (`GDPC1`), civilian employment (`CE160V`) and the civilian labour force (`CLF160V`), the latter two aggregated from monthly to quarterly means — retrieved once from FRED on 2026-08-17 and committed with SHA-256 sums and the retrieval timestamp. The estimation rule (trailing 40-quarter mean of each component, ending at the origin) is recorded in the fixture’s provenance file and was frozen before any forecast was scored. Table 5 shows the decomposition at six origins. Two features are worth reading off. The productivity component is comparatively stable at 1.2–1.5 % per year; the secular movement of g — 2.3 % at the 2010 origin, 1.5 % at 2016, 2.2 % at 2025 — is carried mostly by the labour-force term, and its post-2022 recovery is the immigration surge entering the supply side through nothing more than the origin-local trailing window.

Table 5: The calibrated trend-growth rate g and its decomposition, annualised, at six origins. Each entry is a trailing 40-quarter mean ending at the origin, from measured series only.

Origin	Productivity (%/yr)	Labour force (%/yr)	g (%/yr)
2010Q2	1.53	0.78	2.31
2015Q2	1.16	0.53	1.69
2020Q2	1.43	0.26	1.68
2022Q2	1.24	0.58	1.82
2024Q2	1.47	0.76	2.24
2025Q2	1.41	0.82	2.23

Table 6: Raw-dynamics firewall check at 2024Q4. The right-hand column is the discarded forecast-error-fitted override layer, retained only for contrast; no result in this paper reads it.

Parameter	Stored baseline	Fresh derivation	Discarded fitted override
ρ	0.954811254910	0.954811254910	0.911458333333
r^*	-0.001043926436	-0.001043926436	-0.002333333333
ξ_π	0.789655283206	0.789655283206	0.708333333333
ξ_γ	0.532167779487	0.532167779487	0.258730279487

3.5 The no-fitting firewall

That behavioural parameters have *not* been fitted to forecast errors was verified rather than assumed. The stored baseline carries firewall metadata recording that the parameters are raw, that no forecast-error-fitted parameters are present (`forecast_error_fitted_parameters = false`) and that no post-processing is embedded. Numerically, at 2024Q4 the four Taylor-rule parameters agree to every digit between the stored baseline and a fresh derivation, neither equals a discarded forecast-fitted override layer retained only for contrast, and the two produce bit-identical 12-quarter paths under the same seed (Table 6).

The firewall governs every component of §2 equally. The reconciliation restores an accounting identity; the drift expectation corrects an estimator that is wrong on its own terms; the trend-growth rate is a trailing average of measured series under a rule frozen in advance. None was selected, and none may ever be re-tuned, with reference to a forecast error; re-estimating ω , the Taylor-rule coefficients, g 's window, or the expectation parameters against realised GDP paths would be behavioural tuning, and it is named here explicitly as prohibited. The same discipline binds the DSGE benchmarks of §6.5: their parameters are estimated from pre-origin training data by posterior mode under published priors, never from their forecast performance.

3.6 Absence of behavioural look-ahead

The exogenous arrays for government consumption, exports and imports extend up to twelve quarters past each origin in the artifact. This is behaviourally empty: only the element at the origin index is ever read, once, as a scalar, at construction, after which the series evolve by their own estimated processes. The claim was tested by mutation. Three runs at 2024Q4 under an identical seed — untruncated; truncated to the origin; and untruncated with *every post-origin entry overwritten by -10^{99}* — produced bit-identical real GDP, nominal GDP and policy-rate paths, with the poisoned run finite throughout. Truncation is applied anyway as past-only hygiene, and every estimation window terminates at the origin.

4 From Austria to the United States: what had to change, and why it was hard

The reference exercise is Austrian. Porting it is often described as a data-substitution task. It is not, and the differences explain why two of the model components of §2 exist at all.

4.1 A side-by-side comparison

4.2 Why each difference is hard, not merely different

Valuation bases. A supply–use system can be stated at *basic prices* (what the producer receives) or at *purchasers' prices* (what the buyer pays, including trade and transport margins and net product taxes). The model's coefficients are defined on the basic-price convention; the BEA tables are published at purchasers' prices. Bridging requires netting product taxes

and reallocating margins, and every bridge leaves a residual somewhere. Eurostat hands the Austrian calibration a table that is already on the required basis; the U.S. calibration has to construct one.

The industry/commodity axis. Eurostat publishes symmetric product-by-product tables: rows and columns index the same 64 objects. BEA publishes a *make* table (which industry produces which commodity) and a *use* table (which industry consumes which commodity) over non-identical axes — 71 industries against 68 commodities. Projecting one onto the other is a modelling choice, not a lookup. Put these two paragraphs together and the reconciliation of §3.2 stops looking optional: a published system balances only in the basis and on the axes in which it was published, the bridge and the projection each disturb the balance, and taking imports from the measured T262 matrix removes the residual degree of freedom that could have absorbed the disturbance. For a simulator whose capacity envelope is anchored to those same accounts, the residual is not a bookkeeping blemish; it is a standing order for some sectors to produce beyond their ceiling.

One structural year for fifteen years of origins. The U.S. artifact contains 37 annual arrays of length one — the input–output block, firm counts, employees, population, census unemployed and inactive, fixed assets, dwellings, capital consumption, tax and benefit aggregates, and sectoral wages — frozen at a single reference year. Out of the box only nine origins are constructible. Unlocking the historical grid uses a single-array rewrite: shallow-copy the calibration dictionary and set the year index to the origin’s year, which makes the annual index resolve to 1, the only valid index for a length-one array. All quarterly series run from 1996Q4, so opening *levels* scale with the era (model opening real GDP is 5.483×10^6 at 2019Q4, 3.745×10^6 at 2010Q2, 3.231×10^6 at 2005Q2). The consequence is a clean and declarable split, which the results tables respect:

- **Flow targets** — real GDP growth, deflator inflation, nominal GDP growth, the policy rate — are initialised from quarterly series with full history. The annual structure enters as a fixed technology and demand-composition matrix, not as a level, and §7.8 shows the results are insensitive to which reference year supplies it.
- **Stock-initialised targets** — unemployment above all — open at the structural year’s level at *every* origin. The unemployment *level* is therefore mis-initialised at historical origins by construction; §7.6 scores the target anyway and reports the loss honestly.

5 Scientific assumptions register

Every assumption on which a reported number depends is listed here with its justification and its status. [S] marks a structural choice — one that restores an accounting identity or corrects an estimator that is wrong on its own terms, using past-only information. [T] would mark behavioural tuning against forecast performance; no entry carries it, and none was done.

A1. Revised-data protocol; current-vintage truth, symmetric across models. The scoring panel is a current-vintage snapshot of 101 quarters, 2000Q3–2025Q3, used for estimation and truth by every column in the field. Real GDP growth for 2010Q2 is the value as revised through 2026, not the value available in 2010. *Status*: a known limitation, symmetric across every model, and the same design the reference paper adopts for the bulk of its evaluation. *Flags*: `real_time = false`, `origin_admissible = false`, `promotion_eligible = false`.

- A2. One annual structure at every historical origin.** The 37 frozen annual arrays are future information at origins before the reference year. *Status:* asymmetric and ours alone — the sharpest departure from the reference paper. Bounded for flow targets, and now measured: re-running the exercise under 2017 and 2012 structures moves the headline weighted ratio by at most 0.0007 (§7.8). Remains binding for the opening unemployment level (A10).
- A3. Agent scale 10^{-5} .** 130 firms. *Justification:* an agent-count sweep from 130 to 2708 firms leaves the central results invariant. *Status:* adequate for the mean, *inadequate for dispersion:* 54 of 68 sectors hold exactly one firm at this scale.
- A4. Normal capacity utilisation $\omega = 0.85$ [S, inherited].** Hard-coded upstream. It fixes opening headroom at $1/0.85 - 1 = 17.6\%$ for every firm by construction. *Status:* an inherited magic number. Sourcing it from the Federal Reserve G.17 capacity utilisation series [Board of Governors of the Federal Reserve System, 2025a] would be a structural improvement; sweeping it to improve RMSE would be tuning and is prohibited.
- A5. Opening inventories $S_i(0) = 0$ [S].** As in the reference specification. A signed accounting residual is not an inventory stock; once the balance clears (§3.2) there is no residual to promote.
- A6. Production-account anchoring, $\lambda = 0.983460$ [S].** Stated fully in §3.2: the aggregate excess of uses over supply decomposes exactly into two measurement facts of the source data, one control must move for the reconciliation to be well-posed, and the calibration anchors on the production account. λ is fixed by the accounting identity alone. The alternative closure (hold every budget; let a uniform 1.01 % excess-demand wedge survive) was built and measured; it overshoots (§8.3).
- A7. Random-walk-with-drift growth expectations [S].** The correctly specified estimator for an integrated-with-drift series, on past data only (§2.3, Eq. (2), Table 2). Gated behind a default-off model property, so the Austrian and Italian calibrations remain bit-identical; measured contribution reported in §8.1.
- A8. Trend supply growth at the observed rate [S].** g is a trailing 40-quarter mean of measured productivity and labour-force growth, re-estimated at each origin under a rule frozen in advance (§2.4, §3.4); it advances $\alpha_s, \bar{w}_s, \kappa_s, \delta_s$ jointly, leaving unit labour cost and the replacement share invariant and injecting no demand flow. Book capital remains constant ($K' = K$); capacity growth is carried by the efficiency coefficients, so net $I = \Delta K$ holds trivially. *Status:* an observed-trend closure of a structural absence. The demand-side alternative — investment responding to the capacity gap — was built in two designs and rejected on matched-seed stability diagnostics, never on forecast errors (§8.4).
- A9. No behavioural fitting.** Verified numerically (§3.5); the forecast-error override layer is never read on the paths that produce these results. The DSGE benchmarks are estimated by posterior mode on pre-origin data under published priors (§6.5); their forecast performance never feeds back into their estimation.
- A10. Unemployment is scored, with a declared initialisation defect.** The initial unemployed stock derives from the frozen annual structure, so every historical origin opens at that year’s labour market rather than its own. The target enters the scored tables as its own set; the *level* errors this creates are reported as losses, and the coverage misses are 52–64 % above-band at every horizon — the signature of a level defect, not a dynamics defect (§7.6). For the DSGE columns, which have no unemployment block,

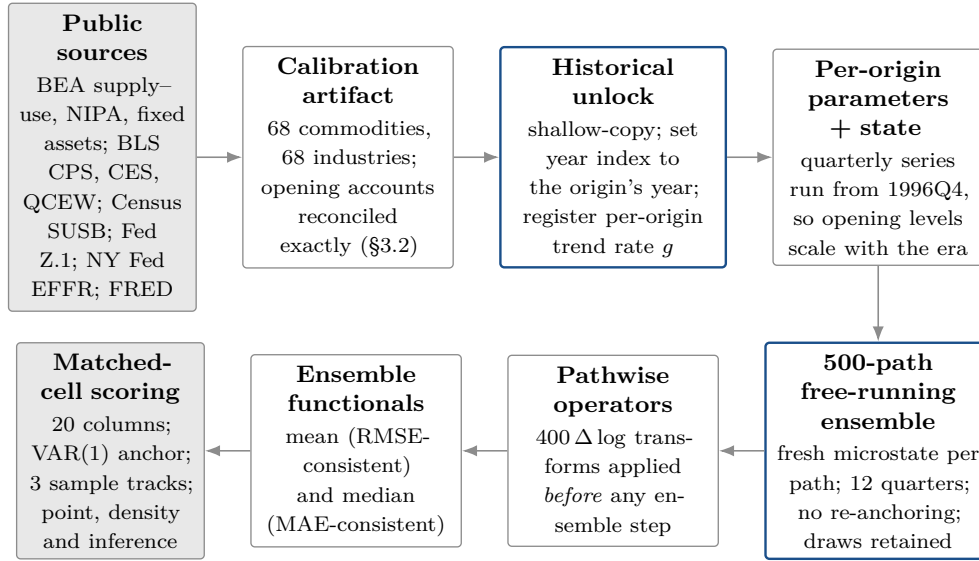


Figure 2: F7 — the forecasting pipeline. Public source data are compiled once into a reconciled calibration artifact. The historical unlock re-points the annual index so the artifact can be instantiated at any origin and registers the origin-local trend rate; the quarterly series, which run from 1996Q4, supply era-appropriate opening levels. Each origin is simulated 500 times, transformed pathwise, reduced to a mean and a median point forecast, and scored — point, density, and inference — on cells shared with every other column.

cells come from a labelled per-origin Okun bridge (§6.5); every table quoting them carries the label.

A11. Seeds and Monte-Carlo policy. Simulation seeds are

$$\text{seed}(\text{origin}, \text{path}) = \text{hash}((:\text{abm_revised_v1}, \text{origin}, \text{path})) \bmod 2^{30},$$

so no two cells share a stream. Every component experiment of §8 draws the *identical* seed stream, which makes each contrast matched-seed; the expectation branch consumes exactly one Normal variate either way precisely to preserve that alignment. The DSGE columns use fixed per-origin seed bases with a domain-separated per-path generator. Failed paths are counted and left visible, never resampled; the count is zero everywhere.

6 Forecast design

6.1 Origins, horizons and the free-running protocol

Figure 2 shows the whole apparatus. Table 8 gives its settings. The protocol — scored columns, calibration rules, target sets, and the success bar of §7.1 — was frozen in a committed document (`scripts/us/forecasting/diagnostics/stage2b/STAGE2B_PROTOCOL.md`) *before* any scored table was produced or examined.

6.2 Operators

Forecast targets are constructed *pathwise, before any ensemble operation*. Writing rg and ng for a path’s real and nominal GDP series and taking row 1 as the origin, so that horizon h is

row $h + 1$:

$$\text{real_gdp} = 400 (\log rg_{h+1} - \log rg_h), \quad (6)$$

$$\text{gdp_deflator} = 400 [(\log ng_{h+1} - \log rg_{h+1}) - (\log ng_h - \log rg_h)], \quad (7)$$

$$\text{nominal_gdp} = 400 (\log ng_{h+1} - \log ng_h), \quad (8)$$

$$\text{EFR} = 100 ((1 + r_{h+1})^4 - 1). \quad (9)$$

The *GDP deflator* is the ratio of nominal to real GDP — the economy-wide price index implied by the accounts — so its log difference is nominal growth minus real growth. The quarterly-to-annual-rate scaling cancels from log differences, so $400 = 4 \times 100$ is the complete annualisation. Unemployment has no field in the data structure and is read off agent state inside the stepping loop, with $H^{\text{act}} = 1\,826$ — the concept closest to the labour-force denominator — in the denominator.

6.3 Ensemble functionals

Two point forecasts are emitted per cell, as separate model columns: the *mean* of the pathwise-transformed values and the *median*. This follows Gneiting [2011]: the optimal point forecast depends on the loss function, the mean being the Bayes act under squared error and the median under absolute error. Reporting both prevents a loss-function mismatch from being mistaken for, or hidden inside, a model property. The full 500-path draw set is additionally retained per cell, which is what makes the exact density scores of §6.6 possible.

6.4 The comparator field: ten statistical columns

Table 9 gives the statistical family, taken from the benchmark module’s own type definitions. All are estimated on training data only at each origin, with hyperparameters fixed before the fit and encoded in the model identifier; none is tuned inside the evaluation.

The anchor is not a strong benchmark — at the 2010Q2 origin its estimated companion matrix has spectral radius 1.0342 and design condition number 8 747, i.e. it is borderline explosive and poorly conditioned — and this should temper any reading of the absolute level of a ratio, though not of the differences between models scored against the same anchor.

A fixed-parameter semi-structural state-space comparator (potential growth, output gap, natural unemployment, neutral real rate, inflation anchor, inflation, policy rate and lagged output gap, updated by an exact Kalman filter) exists in the repository but is not in this field: it cannot produce GDP-deflator forecasts natively, a structural coverage fact recorded in the frozen protocol before any score existed. Its standing from a separate comparison on its own four-target set is quoted as context in the repository reports.

6.5 The comparator field: two DSGE columns

A DSGE model, for the reader who has not met one: a small set of optimising representative agents — a household, a firm sector, a central bank — whose decision rules are derived from explicit intertemporal optimisation under rational expectations, linearised around a balanced-growth steady state. The result is a linear state-space system whose cross-equation restrictions tie today’s observables to expectations of tomorrow’s, solved here by the Sims [2002] *gensys* method and estimated by Bayesian methods: a prior over the structural parameters, a Kalman-filter likelihood over the observables, and (in this work) the posterior *mode* as the parameter estimate at each origin. DSGE models are the institutional standard for structural forecasting at central banks, which is what makes them the benchmark that matters [Smets and Wouters, 2007]. Both columns are re-estimated from scratch at every one of the 61 origins on data

through that origin only; per-origin estimation and solution status is a published artifact; no origin is dropped.

dsge_small_nk.

An An and Schorfheide [2007]-type small New Keynesian model: three observables (real GDP growth, GDP-deflator inflation, the federal funds rate), eight states, three structural shocks — the minimal rational-expectations system with a Phillips curve, an Euler equation and a Taylor rule. The solver mechanics are the repository’s sealed implementation, validated bit-tight against the reference FRBNY DSGE.jl oracle; the column estimates all 13 parameters by posterior mode at every origin on exactly the panel columns the statistical family sees. All 61 estimations converged, the solution is determinate at every origin, and the Taylor principle holds at every mode ($\psi_1 \in [1.31, 1.54]$).

dsge_sw07.

The Smets and Wouters [2007] medium-scale model — the canonical empirical DSGE of the U.S. business cycle: sticky prices and wages, habit formation, investment adjustment costs, variable capacity utilisation, seven structural shocks, and a flexible-price shadow economy that defines the output gap the policy rule responds to. The implementation is a full 49-state transcription of the published replication, solved by a size-generic **gensys** that was oracle-verified to reproduce the repository’s sealed validated solver exactly, and estimated by warm-started posterior mode at every origin under the published priors on the seven standard observables (output, consumption, investment and real-wage growth per capita, hours, deflator inflation, the federal funds rate) from 1966Q1 through the origin. **The SW07 column sees more history and more series than any other column in the field** — an information advantage granted to the challenger deliberately, and disclosed wherever its results are quoted. Its parameter estimates are canonical across all 61 origins (trend growth 0.29–0.42 % per quarter per capita, Taylor coefficient 1.6–2.0, interest-rate smoothing ≈ 0.82); a continuation test at the first origin gained only +2.45 log-posterior from 8 000 further optimisation steps, so the modes are materially converged.

Three conventions travel with every DSGE cell. Their predictive densities (500 paths, matching the ABM) carry filtered-state and future-shock uncertainty at the posterior mode, *without* parameter uncertainty — weaker than the reference paper’s full Bayesian bands, stronger than fixed parameters, and disclosed. Their policy-rate paths are floored at zero (the effective lower bound, which neither model imposes internally). And their unemployment cells come from a labelled per-origin *Okun bridge* — an auxiliary regression of the change in unemployment on real growth, iterated over each model’s own simulated growth paths — because neither DSGE has an unemployment block; the bridge is outside both cores and every table quoting DSGE unemployment carries the label.

6.6 Truth, scores, and inference

Truth is the revised panel itself; the error sign convention is forecast minus actual. Target transformations are *not* uniform across the panel: the four GDP-family series are $400 \ln(\text{level}_t / \text{level}_{t-1})$; the unemployment rate is a quarterly mean of three monthly levels with no transform; EFFR is the equal-weight mean of published business-day rates within the quarter [Federal Reserve Bank of New York, 2025].

Point scores. RMSE (root mean squared error) penalises large errors quadratically; MAE (mean absolute error) linearly; MASE (mean absolute scaled error, Hyndman and Koehler, 2006) divides MAE by the in-sample mean absolute first difference of the same series, making it scale-free. The *weighted ratio* used for standings is a macro-average of matched cellwise ratios,

$$\mathcal{W} = \sum_{\text{targets}} \sum_h w_{\text{target}} w_h \frac{\text{score}_{\text{model}}(\text{target}, h)}{\text{score}_{\text{anchor}}(\text{target}, h)}, \quad w_h \in \{0.30, 0.25, 0.20, 0.15, 0.10\}, \quad (10)$$

with equal target weights and horizon weights declining in h . It is explicitly *not* a ratio of pooled losses, which would let one badly scaled target dominate. The scoring key is (origin, target, horizon), and only cells where *every* column in the field is present survive — 1 415 common cells — so no ranking can be a sample artifact. Three target sets are scored: the headline pair {real GDP growth, GDP-deflator inflation}, the secondary pair {nominal GDP growth, EFR}, and unemployment as its own set.

Density scores. The continuous ranked probability score (CRPS) generalises absolute error to a full predictive distribution: it is the integrated squared difference between the forecast’s cumulative distribution and the step function at the realised value, so it rewards densities that are both *sharp* (narrow) and *correct* (centred), and it reduces to absolute error for a point forecast. It is computed exactly from the retained 500-path draws by the standard pairwise-distance identity. *Interval coverage* asks how often the realised value falls inside a stated predictive interval: a well-calibrated 90 % interval should contain the truth 90 % of the time — more often means the bands are wastefully wide, less often means overconfidence.

Inference. Rankings on 50–61 origins need error bars. Three standard instruments are attached, all on matched cells. The Diebold and Mariano [1995] test asks whether two models’ per-origin loss differences average to zero, with the Harvey et al. [1997] small-sample correction and autocorrelation-robust variance (overlapping multi-quarter forecasts are serially correlated by construction). The Romano and Wolf [2005] stepdown controls the familywise error rate when one model is compared against a whole family at once — it answers “is the ABM better than *each* challenger” without the cherry-picking risk of many unadjusted pairwise tests — via a studentised stationary bootstrap on per-origin weighted losses. The Hansen et al. [2011] model confidence set inverts the question: it trims the field to the subset of models that cannot be statistically distinguished from the best at a given confidence level, so it reports how much the sample can actually separate.

6.7 Sample tracks and the pandemic mask

6.8 Monte-Carlo error

The ABM’s point forecast is an ensemble functional over finitely many stochastic paths, so its measured RMSE is inflated by approximately $\sqrt{1 + (\text{se}_{\text{MC}}/\text{RMSE})^2}$. Table 11 shows that at 500 paths this is negligible: against a real-GDP cross-origin RMSE of order 6 pp, a Monte-Carlo standard error of 0.14 pp inflates the measured RMSE by well under 0.1 %. (At a 16-path pilot count the standard error was 1.3 pp and *was* material; 500 paths were used precisely to remove the confound.)

6.9 The opening-row measurement basis

One artifact of initialisation must be declared because it affects the $h = 1$ cell, which carries the largest horizon weight. The initialiser writes row 1 on a different measurement basis from rows 2 onwards: prices are normalised to 1 at initialisation, so real and nominal GDP coincide in row 1 and the model-implied opening deflator is exactly 1.0. The opening row is the *model-implied* opening, not an observed control — reassuring for leakage, unhelpful for $h = 1$ comparability. Dropping $h = 1$ was rejected outright: it would render the grid incomplete and the model unranked. It is reported and flagged.

Table 7: Structural differences between the reference Austrian calibration and the United States calibration used here. The right-hand column is not a list of approximations to be tightened later; several entries change what the model can be asked to do.

	Austria (reference)	United States (this work)
Input–output source	Eurostat 64-category NACE/CPA product-by-product tables at <i>basic prices</i> , already symmetric and balanced [Eurostat, 2025]	BEA 71-industry $\rightarrow$ 68-commodity make–use system, published at <i>purchasers’ prices</i> , requiring a valuation bridge, an industry $\rightarrow$ commodity projection, and the exact reconciliation of §3.2 [U.S. Bureau of Economic Analysis, 2025]
Firm counts	Business demography: active enterprises per sector	Census Statistics of U.S. Businesses [U.S. Census Bureau, 2025] plus USDA farm counts and pseudo-producers; a mixed enterprise/establishment ontology
Agent scale	Approximately 1:1 with observed entities	10^{-5} : 130 firms, $\approx 2\,831$ modelled people; 54 of 68 sectors hold exactly one firm
Foreign block	Euro-area output and inflation are genuinely <i>exogenous</i> to Austria	The “euro-area” block is filled with <i>U.S.</i> data, so the Taylor rule reacts to an exogenous proxy for the same economy the model is simulating; there is no independent foreign demand state and no policy feedback loop
Trend growth	Low trend growth over a short scored window; frozen supply coefficients are a tolerable abstraction	2%/yr trend with measured productivity and immigration components; frozen supply coefficients collapse the labour market and drain capacity, hence §2.4
Labour and financial accounts	Harmonised Eurostat sector accounts, one vintage, one classification	CPS, CES and QCEW surveys stitched to the Financial Accounts (Z.1) across different statistical universes [U.S. Bureau of Labor Statistics, 2025, Board of Governors of the Federal Reserve System, 2025b]
Unemployment insurance	A single national replacement rate	State-level programmes varying by wage, duration and emergency extension; the Austrian constant $\theta^{UB} = 0.55(1 - \tau^{INC})(1 - \tau^{SIW})$ survives unchanged in the code
Trade closure	Imports read off the balanced table; the balance clears by construction	Imports measured from BEA T262 while the supply–use residual is <i>not</i> re-absorbed; the residual is closed exactly by the RAS reconciliation
Structural vintage	Structure recalculated at or near each forecast origin	One 2024 structural row carried backwards to all 61 origins; measured directly in §7.8 by re-running the exercise under 2017 and 2012 structures

Table 8: Experimental design. “Free-running” means the model is initialised once at the origin and stepped forward twelve quarters with no re-anchoring, no observed-data injection and no conditioning on realised post-origin values. This corresponds to the reference paper’s unconditional forecast, the harder of its two exercises.

Origins	61 (2010Q2–2025Q2)	Paths per origin	500
Minimum training window	40 quarters	Path failures	0
Horizons scored	1, 2, 4, 8, 12	Simulation mode	free-running, serial
Simulated quarters per path	12	Model scale	10^{-5}
Burn-in quarters	0	Julia / threads / BLAS	1.10.3 / 1 / 1

Table 9: The ten statistical benchmarks. The VAR(1) is the ratio anchor: every reported ratio is a model’s score divided by the VAR(1)’s score on the same cells.

#	model_id	Specification
1	naive_no_change	Random walk: forecast equals the last training observation.
2	naive_drift	Random walk with drift estimated from training differences only.
3	naive_historical_mean	Constant forecast equal to the training-sample mean.
4	univariate_ar_p1_constant	Univariate AR(1) with intercept, per target.
5	univariate_ar_p4_constant	Univariate AR(4) with intercept.
6	univariate_ar_bic...	Lag order chosen by the Bayesian information criterion on a common, training-only window.
7	beforeit_var_p1_constant	Multivariate VAR(1) with intercept over the eight-target panel. Ratio anchor.
8	beforeit_var_p2_constant	VAR(2).
9	beforeit_var_p3_constant	VAR(3).
10	bvar_mniw_v1_p1...	Natural-conjugate Bayesian VAR with a matrix-normal / inverse-Wishart prior and Minnesota-style shrinkage [Litterman, 1986]: tightness 0.2, lag decay 1.0, own-lag prior mean 0.0, intercept variance 100, inverse-Wishart degrees-of-freedom offset 2, innovation scale 1.0, scale floor 10^{-8} .

Table 10: The three sample tracks and their common-cell counts. The pandemic mask is applied to *target* periods, not origins, and was pre-registered in the run manifest rather than selected after seeing results.

Track	Rule	n at $h = 1/2/4/8/12$
all-available	every common cell	61 / 60 / 58 / 54 / 50
balanced $h=12$	origins with a complete 12-quarter horizon	50 / 50 / 50 / 50 / 50
pandemic-masked	exclude cells whose target period falls in 2020Q1–2021Q4	53 / 52 / 50 / 46 / 42

Table 11: Monte-Carlo error at 500 paths, ranges over the five horizons, from the model’s own run. “Ensemble s.d.” is the dispersion of the simulated paths — genuine model uncertainty; “MC s.e.” is the sampling error of the ensemble mean — simulation noise.

Target	Mean ensemble s.d.	Mean MC s.e. of the mean	Max MC s.e.
real_gdp	3.017–3.274	0.135–0.146	0.249
gdp_deflator	0.860–1.079	0.038–0.048	0.096
nominal_gdp	3.184–3.460	0.142–0.155	0.258
unemployment_rate	0.345–1.418	0.015–0.063	0.090
effective_federal_funds_rate	0.139–0.337	0.006–0.015	0.018

7 Results

7.1 The preregistered bar, and the verdict stated first

The frozen protocol committed to a success bar before any scored table existed: the ABM is superior if its weighted headline-pair RMSE ratio beats the best non-ABM column, *including both DSGE columns*, in at least two of the three tracks with Diebold–Mariano significance reported per horizon, *and* its density scores (CRPS, 90 % coverage) are no worse than the best challenger’s; if it instead wins only part of that, the part it wins is the claim published; and a negative result is published with the same tables.

The verdict against that bar. The ABM takes the **best weighted headline-pair RMSE ratio of all twenty columns in all three tracks** — 0.828 / 0.828 / 0.792 (mean column; the median column is marginally better at 0.827 / 0.827 / 0.791) — against 0.857 / 0.856 for the best non-ABM column in the two full-sample tracks (SW07) and 0.848 in the pandemic-masked track (the AR(1)). The point condition is met in three of three tracks, not merely the required two. Romano–Wolf stepdown rejects equal accuracy against *both* DSGE columns at the 5 % level in every track (adjusted $p = 0.0002$ – 0.016); against the statistical family the weighted-loss test is marginal (adjusted $p \approx 0.08$). The ABM’s CRPS beats the best challenger in all three tracks. But its **90 % interval coverage fails the “no worse than the best challenger” clause in the two full-sample tracks** — pooled headline coverage 0.832 / 0.812 / 0.906, against a challenger within 0.005–0.008 of nominal — and the miss is located: real-GDP coverage is 0.86–0.95 at every horizon, while the GDP-deflator bands are too narrow (0.70–0.79). **The full bar is therefore not met**, and per its honest-success clause the claim this paper publishes is the conditional one:

On revised data, at the reference paper’s own evidentiary standard, the ABM forecasts the headline pair {real GDP growth, GDP-deflator inflation} *more accurately in point terms than every statistical and equilibrium challenger in the field, in all three tracks* — the margin over both DSGE columns is significant at 5 % under multiplicity control, and the real-GDP margin over SW07 is significant at business-cycle horizons — *and more sharply in CRPS terms*, while its GDP-deflator intervals are under-dispersed relative to the best-calibrated challenger. The full preregistered bar fails only on that interval-calibration leg.

7.2 Standings

Table 12 and Figure 3 give the standings; ranks 3–4 in each track are the ABM’s two component-ablation columns (§8), which are excluded from the table to keep it a field comparison. Three readings.

The strongest challenger is the DSGE, and the ABM beats it. SW07 at 0.857 / 0.856 beats every statistical model in both full-sample tracks — consistent with the DSGE-forecasting literature, in which estimated medium-scale DSGE models are competitive with Bayesian VARs — so the ABM’s margin over it is a margin over the strongest available challenger, not over a straw man. The margin is 0.029–0.030 weighted-ratio points in the full-sample tracks, and §7.7 shows it survives multiplicity control.

The pandemic mask reorders the field, and the ABM’s margin widens. Masking 2020Q1–2021Q4 target cells drops SW07 to 0.954 (10th) and the small-NK to 1.134 (14th) — both equilibrium columns owe part of their full-sample standing to the pandemic cells, where their mean-reverting structure absorbed the rebound — while the AR(1) becomes the best non-ABM column at 0.848. The ABM is first at 0.791–0.792, its *widest* margin of the three tracks: its accuracy does not depend on the extreme cells.

The small NK model is mid-field. Three observables and thirteen parameters are enough to beat the anchor (0.890 / 0.885) but not the historical-mean constant, and its long-horizon

Table 12: Headline standings, {real GDP growth, GDP-deflator inflation}: weighted RMSE ratios against the VAR(1) anchor, rank out of the 20 scored columns in parentheses. Selected rows; the full twenty-column tables, including MAE and the secondary and labour target sets, are the committed artifact output/us_forecasting/stage2b_scorecard/STAGE2B_TABLES.md. Revised-data, mixed-vintage diagnostic.

Column	all-available	balanced $h=12$	pandemic-masked
ABM, median	0.827 (1)	0.827 (1)	0.791 (1)
ABM, mean	0.828 (2)	0.828 (2)	0.792 (2)
dsge_sw07	0.857 (5)	0.856 (6)	0.954 (10)
naive_historical_mean	0.879 (7)	0.884 (7)	0.881 (8)
dsge_small_nk	0.890 (9)	0.885 (9)	1.134 (14)
Minnesota BVAR	0.908 (12)	0.909 (12)	0.852 (6)
univariate_ar_p1_constant	0.918 (13)	0.918 (13)	0.848 (5)
beforeit_var_p1_constant (anchor)	1.000 (16)	1.000 (16)	1.000 (12)

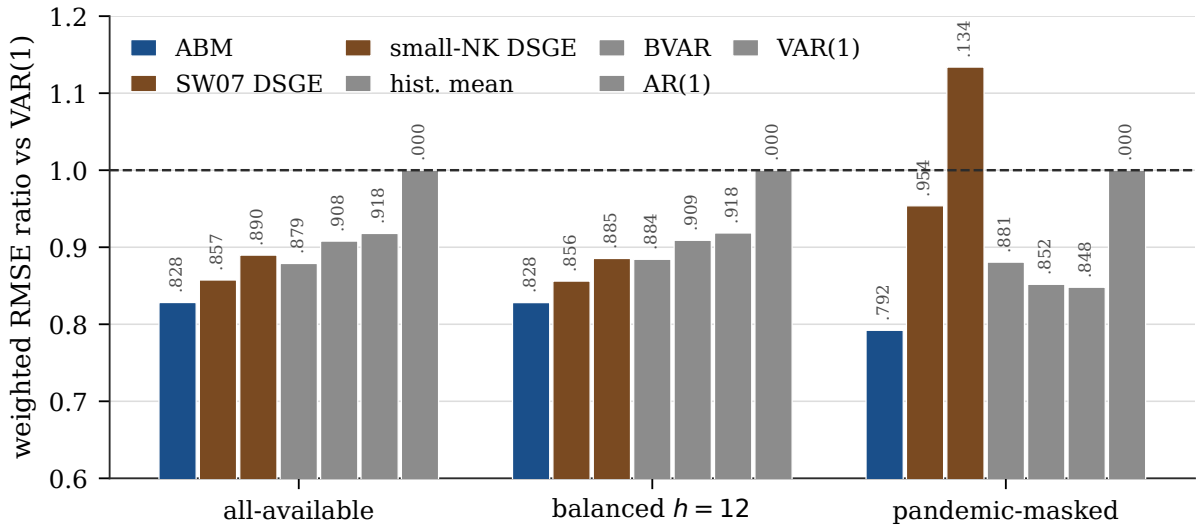


Figure 3: F1 — weighted RMSE ratio on the headline target pair, three sample tracks, selected columns. Blue is the ABM (mean column); dark grey the two DSGE columns; light greys the statistical family. The dashed line is the VAR(1) anchor at 1.0. The ABM is first in all three tracks, and its margin is widest in the pandemic-masked sample. Data: stage2b_weighted_scores.csv.

forecasts revert to a posterior-mode trend that the 2000s-only panel identifies weakly — a known small-sample property of the class, reported as measured.

7.3 Per-target, per-horizon detail

Table 13 and Figure 4 carry the detail. Real-GDP bias is below a quarter of a percentage point in magnitude at every horizon — slightly negative at short horizons, slightly positive at $h \geq 8$ where the trend-growth component holds the profile up — and the level of real-GDP RMSE, about 6 pp against quarter-on-quarter annualised growth whose cross-origin dispersion includes the pandemic, is worth internalising: *nobody* forecasts these series well in absolute terms; the contest is over ratios. The deflator is accurate in ratio terms (0.77–0.97) with a persistent mild under-forecast that grows to -0.6 pp at $h = 12$; its rank oscillates because half a dozen columns sit within a few percent of each other on this target. Nominal GDP — the sum of the two headline components — is first or second at four of five horizons. The policy rate inherits the model’s Taylor rule reacting to a corrected output path; its increasingly negative long-horizon bias (-0.8 pp at $h = 12$) reflects rate paths that ease toward neutral faster than the sample did.

Table 13: Per-target, per-horizon detail for the ABM (mean column), *all-available* track. Ratios are against the VAR(1) anchor; ranks are out of the 20 scored columns. Bias is the mean error (forecast – realised) in annualised percentage points; unemployment bias is in percentage points of the rate.

Target	h	n	RMSE	ratio	rank	bias
real_gdp	1	61	6.088	0.670	1/20	−0.03
real_gdp	2	60	6.113	0.832	3/20	−0.04
real_gdp	4	58	6.197	0.976	2/20	−0.02
real_gdp	8	54	6.421	0.817	2/20	+0.23
real_gdp	12	50	6.652	0.988	4/20	+0.25
gdp_deflator	1	61	1.429	0.794	8/20	−0.19
gdp_deflator	2	60	1.655	0.774	14/20	−0.29
gdp_deflator	4	58	1.934	0.820	4/20	−0.40
gdp_deflator	8	54	2.035	0.973	4/20	−0.53
gdp_deflator	12	50	2.116	0.894	8/20	−0.62
nominal_gdp	1	61	6.839	0.656	1/20	−0.22
nominal_gdp	2	60	6.812	0.781	4/20	−0.33
nominal_gdp	4	58	6.991	0.947	2/20	−0.42
nominal_gdp	8	54	7.279	0.833	2/20	−0.29
nominal_gdp	12	50	7.535	0.973	1/20	−0.37
EFFR	1	61	0.373	0.563	7/20	−0.04
EFFR	2	60	0.693	0.557	8/20	−0.09
EFFR	4	58	1.219	0.588	6/20	−0.20
EFFR	8	54	1.891	0.795	3/20	−0.50
EFFR	12	50	2.164	0.805	4/20	−0.79
unemployment_rate	1	61	2.829	1.345	16/20	−1.91
unemployment_rate	2	60	2.765	0.836	14/20	−1.87
unemployment_rate	4	58	2.653	0.637	11/20	−1.84
unemployment_rate	8	54	2.672	1.192	9/20	−2.00
unemployment_rate	12	50	2.962	0.760	10/20	−2.36

Unemployment is discussed in §7.6.

7.4 Densities: the sharpest CRPS in the field, one calibration miss

Table 14 and Figure 5 give the density evidence, and the trade it reveals is instructive. On real GDP the ABM dominates: sharper by CRPS than SW07 by 17–31 % per track *and* well calibrated (coverage 0.91–0.98 against nominal 0.90). On the deflator the ranking splits by criterion: the ABM’s CRPS is competitive (better than SW07 in the masked track, marginally worse in the full tracks) but its bands are too narrow — coverage 0.70–0.79 pooled, 0.70–0.79 by horizon — while the DSGE columns reach near-nominal coverage through intervals roughly 30 % wider, which is precisely the width their CRPS pays for. Combined over the pair the ABM leads CRPS in every track (1.58 / 1.67 / 0.88 against SW07’s 1.80 / 1.91 / 1.12), and fails the coverage clause of the bar in the two full-sample tracks. The model’s inflation ensemble is genuinely under-dispersed — a mechanism-level fact (the price block inherits little of the quantity dispersion), measured here and named in §10. The policy-rate bands are far too tight at long horizons (coverage 0.16 at $h = 12$), a known Taylor-block limitation now quantified.

7.5 Secondary targets

On the secondary pair {nominal GDP growth, EFFR} the ABM’s weighted ratios are 0.715 / 0.702 / 0.736 (mean; the median column 0.714 / 0.702 / 0.737): first and second in the two full-sample tracks, second and third in the pandemic-masked track behind the AR(4) at 0.719.

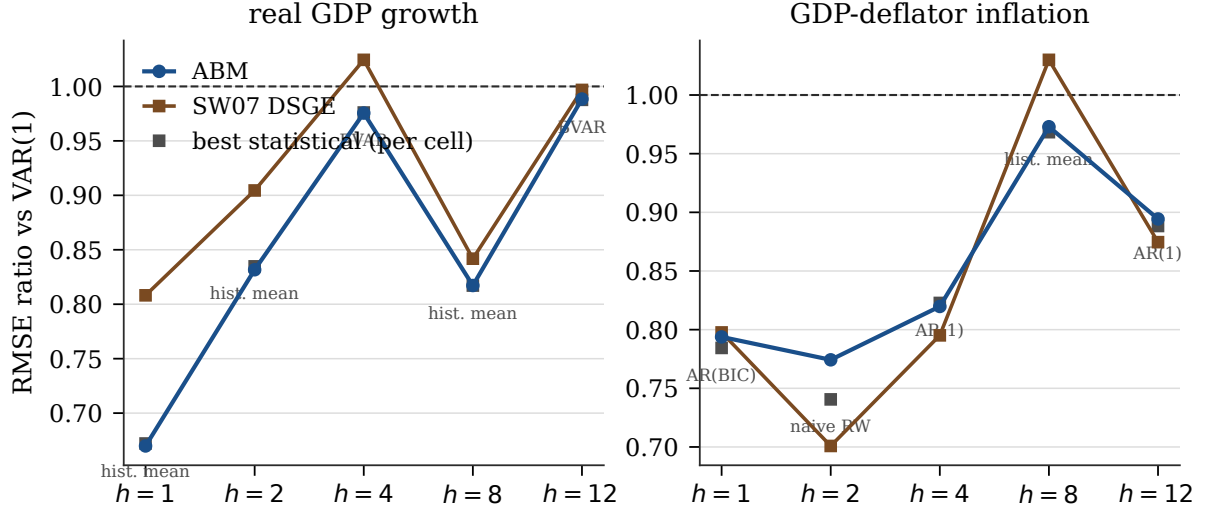


Figure 4: F2 — RMSE ratio against horizon, all-available track. Left: real GDP growth. Right: GDP-deflator inflation. Blue is the ABM; dark grey SW07; grey squares mark the best statistical column *in each cell separately*, with its identity annotated — that envelope is a stiffer comparison than any single benchmark. Data: `stage2b_relative_scores.csv`.

Table 14: Density scores on the headline targets, per track: horizon-weighted exact-ensemble CRPS (lower is better) and 90 % central-interval coverage pooled across horizons (nominal 0.90). ABM and DSGE columns only — the frozen statistical registry is point-only. DSGE densities carry filtered-state and future-shock uncertainty at the posterior mode, without parameter uncertainty. Data: `output/us_forecasting/stage2b_scorecard/STAGE2B_TABLES.md`.

Column	all-available		balanced $h=12$		pandemic-masked	
	real GDP	deflator	real GDP	deflator	real GDP	deflator
<i>Weighted CRPS</i>						
ABM	2.234	0.926	2.365	0.977	1.054	0.713
<code>dsge_small_nk</code>	3.111	0.977	3.353	1.012	1.946	0.772
<code>dsge_sw07</code>	2.697	0.894	2.895	0.933	1.518	0.722
<i>90 % interval coverage (nominal 0.90)</i>						
ABM	0.916	0.748	0.908	0.716	0.982	0.830
<code>dsge_small_nk</code>	0.929	0.826	0.920	0.804	0.981	0.913
<code>dsge_sw07</code>	0.933	0.877	0.924	0.860	0.988	0.940

The masked-track loss is reported as a loss. The policy-rate operator maps the model’s internal Taylor-rule rate to an annualised percentage point; this is explicitly not an approved EFR bridge, and the caveat travels with every number quoting it.

7.6 Unemployment: scored, and lost on the level

Unemployment enters the scored tables as its own target set, and the ABM does not win it: at VAR(1) parity on the weighted ratio unmasked (0.995 mean / 0.974 median, 12th–13th of 20) against persistence models at 0.65, and far behind in the masked track (5.3–5.4, where masking removes exactly the cells in which the VAR anchor itself exploded, deflating every level-biased column’s denominator). The reason is measured and initialisation-shaped, not dynamic: every origin opens at the frozen structural year’s labour market rather than its own, the bias column of Table 13 is a near-constant -1.9 to -2.4 pp, and the interval misses are 52–64 % *above-band* at every horizon — realised unemployment sitting above bands that opened too low. The model’s unemployment *dynamics*, by contrast, now behave: ensemble dispersion

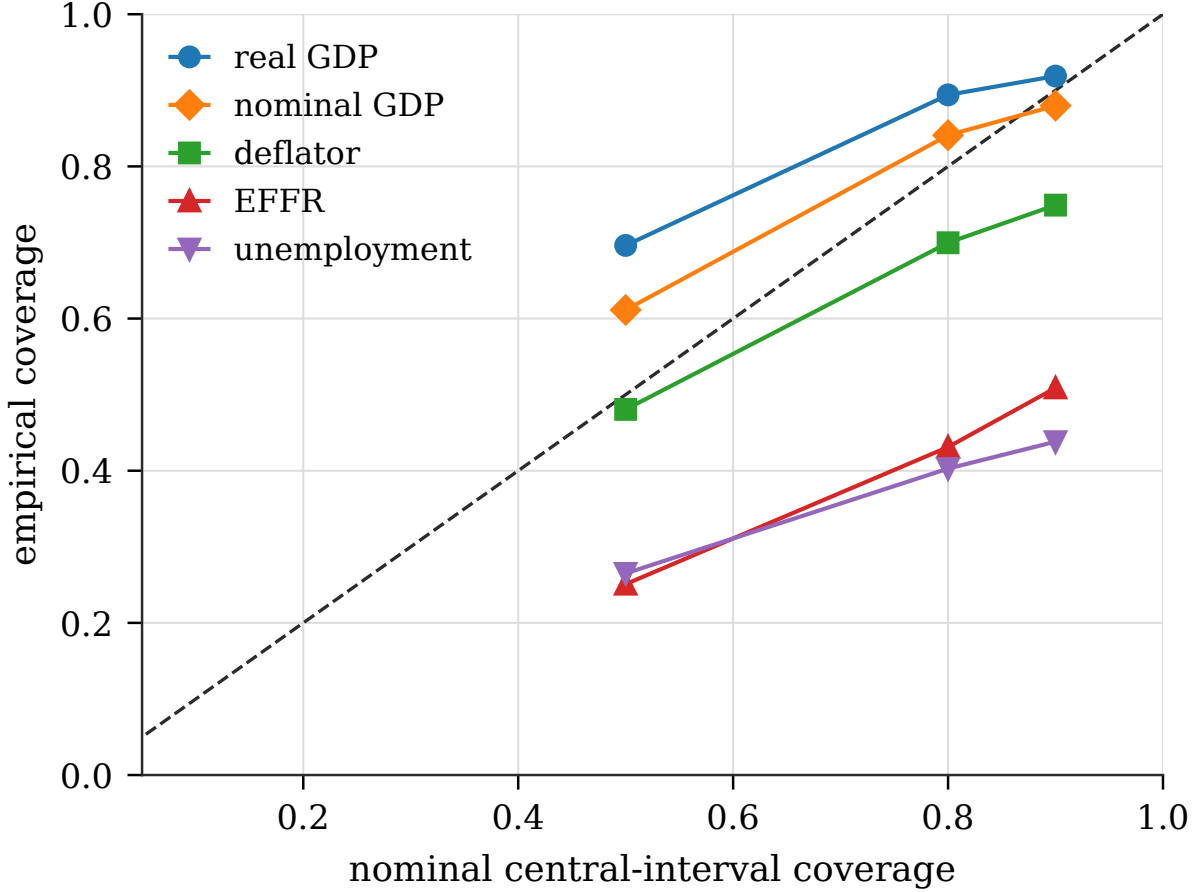


Figure 5: F4 — empirical against nominal interval coverage, all-available track, pooled over the five scored horizons. Points on the dashed 45° line are perfectly calibrated; below the line is over-confident. Real and nominal GDP sit on the line at the 80 % and 90 % levels; the deflator and the policy rate are under-dispersed at every level. Data: `stage2b_density_coverage.csv`.

risers monotonically with horizon to 1.42 pp at $h = 12$, which is $0.83\times$ the realised cross-origin dispersion (Figure 6) — and §8.3 shows by ablation that without the trend-growth component the same statistic collapses to $0.27\times$ as every path exhausts the labour pool. Scoring the level honestly requires origin-correct opening labour stocks — a data acquisition, not a model change, and it is named in §10.

7.7 Formal inference

Table 15 carries the two headline panels. Panel A: Romano–Wolf rejects equal weighted-loss accuracy against all four DSGE columns at the 5 % level in every track (the adjusted p -values span 0.0002–0.016), while against the statistical family the weighted-loss test is marginal (adjusted $p \approx 0.08$ in the full-sample tracks; the historical-mean constant is the loss-closest statistical column through the pandemic cells). Panel B: the real-GDP margin over SW07 is significant at $h = 2, 4, 8$ in every track, insignificant at $h = 1$ in the full-sample tracks — where pandemic-origin outliers inflate the differential variance — and at $h = 12$; the GDP-deflator margin is statistically indistinguishable from SW07 at every horizon (all two-sided $p \geq 0.26$). The ABM’s headline advantage over the strongest DSGE is, precisely, a *real-activity advantage at business-cycle horizons*.

The model confidence set completes the picture as a power statement. In the two full-sample tracks the ABM (mean) is the set’s best member with MCS $p = 1.0$ and no column is excluded at $\alpha = 0.10$ — 50–61 weighted observations with pandemic outliers cannot separate twenty

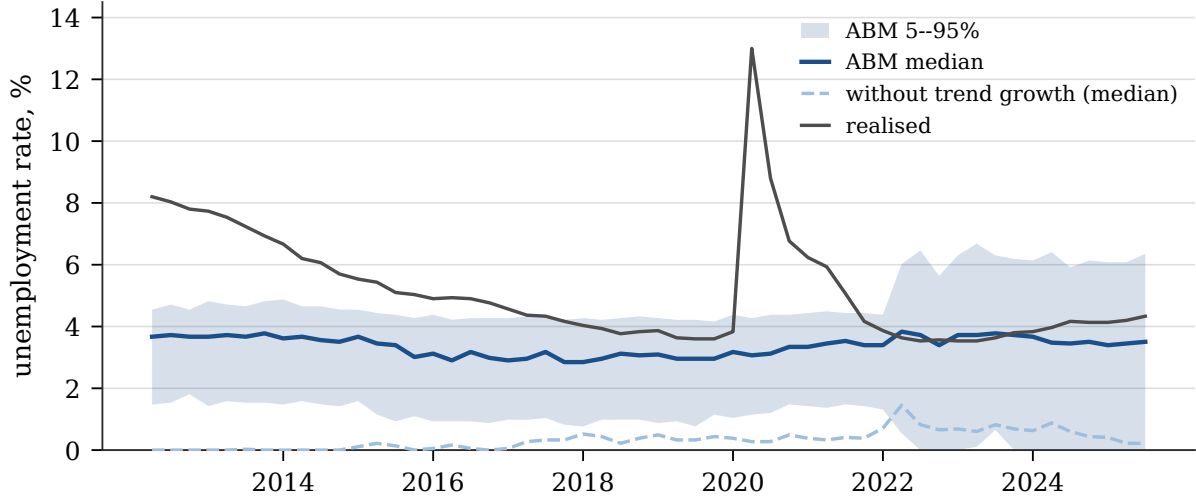


Figure 6: F8 — unemployment at $h = 8$ across the backtest. The ABM’s ensemble median and 5–95 % band against realised unemployment, with the no-trend-growth ablation’s median for contrast. The ablation slides toward the floor at every origin — the labour-pool exhaustion of §2.4 — while the model holds a stable band whose dispersion grows with horizon. The level offset at the recession origins is the frozen opening stock of A10. Data: `abm_ensemble_summaries.csv` (both variants), `quarterly_panel.csv`.

columns. The masked track excludes both small-NK columns, both no-reconciliation ablation columns and `naive_drift`; the ABM leads the surviving set.

7.8 Structural-vintage robustness

The single-structural-year asymmetry of assumption A2 is measured directly. Two additional annual structures were built from revised BEA supply–use tables for reference years 2017 and 2012 through the identical ingestion path (golden-tested: the same code rebuilds the 2024 artifact bit-exactly), identically reconciled (RAS exactly as in §3.2; $\lambda = 0.98306$ and 0.97911 against 2024’s 0.98346 ; residual decompositions published beside the artifacts), and verified; the ABM was re-run at 500 paths on all 61 origins under each, with matched seeds, against the identical field — whose statistical and DSGE columns do not depend on the annual structure and therefore serve as invariance controls.

The headline conclusion does not depend on the structural vintage (Table 16). This also bounds the mixed-vintage worry for the technology matrix specifically: if the frozen recent structure were doing unfair work at historical origins, replacing it with the structure those origins actually had access to would show it; it does not. (Counts, taxes and the opening labour stock remain tied to the structural year — the unemployment level defect of §7.6 stands.)

Table 15: Inference on the headline pair. Panel A: Romano–Wolf stepdown adjusted p -values, ABM against each DSGE column, on per-origin weighted headline losses over complete origins; all twelve cells reject equal accuracy at the 5% level (entries below the rendering precision of the repository tables are shown as < 0.001 ; the smallest adjusted p in the family is 0.0002). Panel B: two-sided HLN Diebold–Mariano p -values, ABM (mean) against `dsge_sw07`, real GDP growth, by horizon. Data: `output/us_forecasting/stage2b_scorecard/STAGE2B_TABLES.md`.

Panel A — Romano–Wolf adjusted p , ABM vs the DSGE columns

Comparator	all-available	balanced $h=12$	pandemic-masked
dsge_small_nk	< 0.001	< 0.001	0.010
dsge_small_nk_median	< 0.001	< 0.001	0.010
dsge_sw07	0.016	0.016	< 0.001
dsge_sw07_median	0.016	0.016	< 0.001

Panel B — HLN-DM two-sided p , ABM vs dsge_sw07, real GDP growth

Track	$h = 1$	$h = 2$	$h = 4$	$h = 8$	$h = 12$
all-available	0.296	0.001	0.022	0.003	0.153
balanced $h=12$	0.301	0.001	0.021	0.004	0.153
pandemic-masked	0.002	< 0.001	0.002	< 0.001	0.104

Table 16: Headline weighted RMSE ratio of the ABM (mean column) under three annual structures, identical field, matched seeds and cells. The maximum deviation, 0.0007, is an order of magnitude below the ± 0.005 Monte-Carlo error of a 500-path ensemble.

Annual structure	all-available	balanced $h=12$	pandemic-masked	headline rank
2024 (main run)	0.8282	0.8281	0.7921	1–2 of 20
2017	0.8281	0.8278	0.7933	1–2 of 20
2012	0.8288	0.8287	0.7922	1–2 of 20

8 Which components carry the accuracy: matched-seed evidence

A model that wins a horse race invites one question above all: *which part of it is doing the work?* Because every component of §2 is flag-gated, each can be switched off and the entire 61-origin exercise re-run under the identical seed stream — every path draws the same random numbers, so differences are the component’s causal effect, not Monte-Carlo noise — and scored on the identical common cells. Two rejected designs are reported alongside, because a design space is only credible if it contains something that failed.

8.1 Attribution

Table 17 orders the mechanisms. The *accounting reconciliation carries the bulk of the accuracy*: without it the headline ratio deteriorates by 0.065–0.075 in the full-sample tracks and by 0.476 masked — the unreconciled model’s standing depended on the pandemic cells. The *expectation specification carries most of the rest*, and matters precisely where the pandemic cells are removed ($+0.061$ masked): the log-level AR(1)’s trend under-delivery (Table 2) binds in calm samples, while in the full sample the 2020 collapse rewards any downward-biased forecast. The *trend-growth component is deliberately headline-neutral* (± 0.004): its payload is the labour market and long-run validity (§8.3), and it did not purchase headline accuracy at their expense — the one interaction worth recording is that under AR(1) expectations the trend-growth economy’s unemployment drifts *up* (to 4.7% at $h = 12$ from a 3.7% opening at the 2010Q2 origin), exactly as balanced-growth accounting predicts when demand expectations under-deliver the trend that

Table 17: Matched-seed, identical-cell component attribution. Effects are changes in the weighted headline RMSE ratio (all-available / balanced / pandemic-masked); negative is an improvement. Each ablation column is itself fully scored in the field artifact.

Experiment	Isolates	Weighted headline effect
Without the opening-balance reconciliation (raw accounts, log-level AR(1) expectations)	the accounting component of §3.2	+0.075 / +0.065 / +0.476 against the full model (0.905 / 0.894 / 1.272: mid-field, and pandemic-dependent)
Log-level AR(1) expectations under the full model	the expectation specification of §2.3	+0.006 / +0.006 / +0.061 (0.835 / 0.834 / 0.853)
Without trend supply growth	the balanced-growth component of §2.4	+0.002 / +0.001 / +0.004 on the headline — <i>and</i> unemployment dynamics degenerate (§8.3)
Demand-side investment feedback (two designs)	the alternative to §2.4	rejected on stability diagnostics before scoring (§8.4)

supply now provides.

8.2 What the unreconciled accounts do to the economy

The unreconciled ablation is not merely worse; it is worse in a way that identifies the mechanism, which is why the reconciliation can be called a component rather than a patch. Unreconciled, the model under-forecasts real GDP growth by -1.3 to -2.9 pp at every horizon (and -7.3 pp at $h = 2$; Figure 7). Matched-seed decomposition attributes 76 % of the average shortfall to the opening-balance defect colliding with the frozen capacity envelope — 12.9–14.4 % of gross output produced under a binding ceiling every quarter, exports filling only 87 % of demand, over-demanded sectors pinned at their caps while under-demanded capacity idles uselessly — and 24 % to the expectation shrinkage of Eq. (2). The diagnosis made a falsifiable prediction: a *quantity-rationing* defect should leave the *price* block untouched when repaired. It held — the GDP-deflator column is bit-identical at $h = 1$ between the two runs and its ratios differ in the third decimal elsewhere, while real-GDP bias falls by two orders of magnitude, to $|\cdot| \leq 0.25$ pp (Table 13). A dose–response check completes the identification: relieving the capacity envelope by +30 % in the unreconciled economy recovers most of the lost growth, while the same relief added *on top of* the reconciliation moves nothing — the two interventions are substitutes for one binding constraint, which is what the mechanism claims.

8.3 What trend supply growth buys: the two gates

Two simulation-internal gates certify the trend-growth component; neither consulted a forecast error.

Utilisation and capacity. Matched-seed 24-quarter simulations at three origins — twice the scored window, because the question is asymptotic behaviour: without the component, utilisation drifts from its 0.850 opening to 0.98–0.99 by quarter 24 with zero capacity growth; the economy grows only by draining its fixed 17.6 % headroom, and beyond roughly eight to ten years it must stall. With the component, utilisation stabilises at 0.87–0.91 while capacity grows 10–13 % over the six years — the calibrated trend. Growth is no longer headroom drainage, and the validity restriction that would otherwise cap the model at $h \leq 12$ on the supply side is lifted. (The rejected alternative closure of assumption A6 — keep every measured budget and let a uniform 1 % excess-demand wedge survive — fails this same gate from the other side: it exhausts its headroom in about four years, which is why its smaller measured headline bias was never taken as evidence in its favour.)

Unemployment dynamics. Without the component, simulated demand growth must be met by employment growth out of the fixed pool: unemployment falls mechanically at every origin,

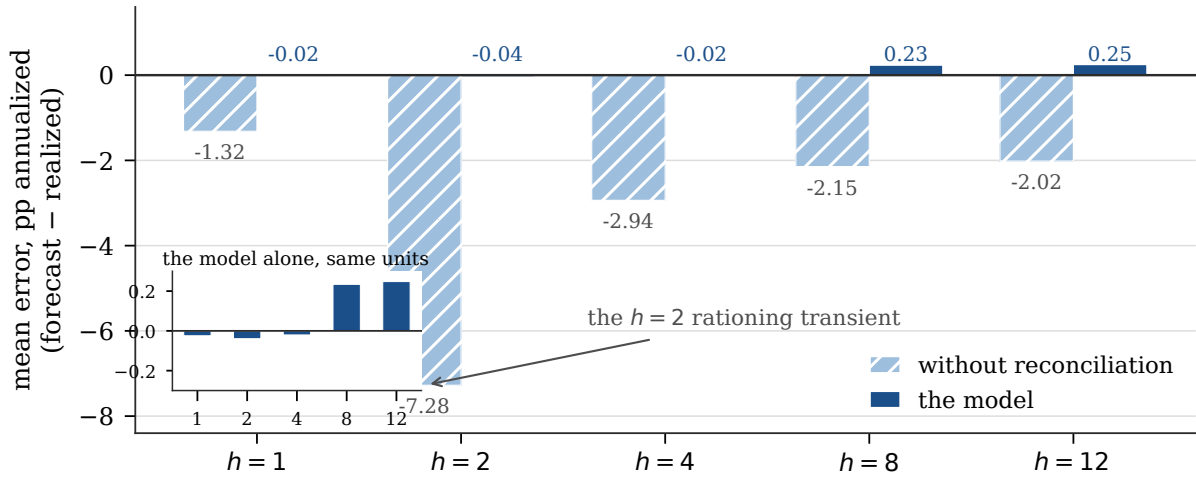


Figure 7: F3 — real-GDP mean error by horizon: the accounting component’s causal effect. Matched seeds, identical cells. Without the reconciliation the model under-forecasts at every horizon, with a -7.3 pp transient at $h = 2$; the full model’s bars are barely visible at this scale (inset: largest $|\text{bias}|$ is 0.25 pp). Data: `stage2b_score_summaries.csv`.

path dispersion collapses to $0.27\times$ the realised cross-origin dispersion by $h = 12$ as every path reaches the floor, and end-of-horizon unemployment sits at $0.8\text{--}1.1\%$ — roughly a third of the lowest rate the United States has recorded in sixty years. With the component, dispersion rises monotonically to $0.83\times$ realised at $h = 12$ and the rate holds a stable, widening band (Figure 6). The remaining coverage misses are $52\text{--}64\%$ above-band at every horizon: the frozen opening *level* of A10, not the dynamics.

8.4 The investment designs that were rejected

The natural objection to §2.4 is that capacity growth should come from *investment* — firms observing demand pressure and accumulating capital — rather than from an exogenous efficiency trend. Two such designs were built, run matched-seed, and rejected on stability diagnostics alone, before any forecast was scored; both remain runnable in the repository. A *capacity-gap accelerator* (I^d closing a fraction $\nu = 0.0625$ of the gap between demand-implied and actual capital each quarter) and a *balanced-growth net-investment flow* ($g K_i$ added to desired investment) each destabilise the economy the same way: the injected investment demand lands in a rationed goods market, raises realised sales, which raises adaptive sales expectations, which raises next quarter’s demand — a multiplier–accelerator spiral that the model’s decentralised markets amplify rather than damp. Measured on matched seeds: real growth $5.5\text{--}8.2\%$ annualised and unemployment to zero within 4–8 quarters, under either design. That fragility is itself a finding about this model class at this calibration: the demand side cannot absorb a net-investment flow without a boom, so trend capacity growth has to enter on the supply side — which is also where, in the data, most of it lives (measured productivity growth). A demand-side investment block stable enough to carry trend growth is an open design problem, not a small parameter choice.

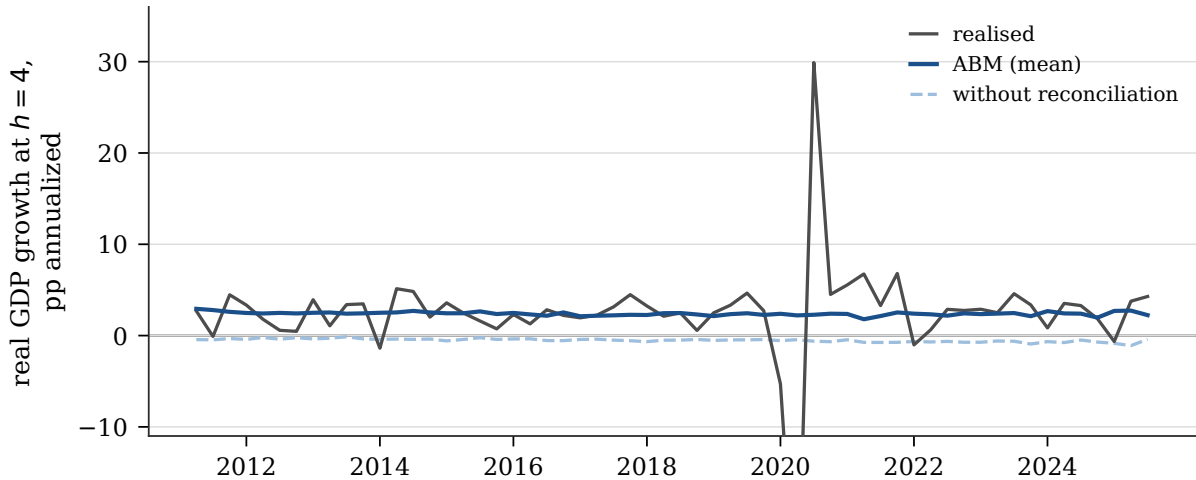


Figure 8: F9 — four-quarter-ahead real GDP growth across the whole backtest. Each point on the model curves is the ensemble functional from a different origin. The unreconciled ablation sits at approximately zero growth throughout; the full model sits near +2.3% and tracks the level of realised growth. Neither forecasts the pandemic quarters, which is why the pandemic-masked track exists. Data: `abm_ensemble_summaries.csv` (both variants), `quarterly_panel.csv`.

9 The current outlook — unscored

Two origins beyond the end of the revised panel — 2025Q4 and 2026Q1 — were run with 500 paths and zero path failures. There is no realised truth for these, so **nothing in this section is scored** (`scored = false`, `realized_truth_available = false`).

From the 2025Q4 origin, average real growth over $h = 1 \dots 4$ is +2.20% with a 5th–95th percentile band of roughly -5.4 to $+9.9$ at $h = 1$; deflator inflation runs 3.21% at $h = 1$, easing to 2.54% by $h = 4$ and $\approx 2.3\%$ by $h = 12$; the policy rate declines from 3.86% to 3.34% by $h = 12$; and unemployment holds 3.7–3.9% over the first year, easing to 3.3% by $h = 12$. From the 2026Q1 origin, average real growth over $h = 1 \dots 4$ is +2.31%, deflator inflation eases from 3.13% to 2.49% by $h = 4$, the policy rate declines from 3.60% to 3.11% by $h = 12$, and unemployment holds 3.4–3.8% throughout. Figure 9 shows the 2026Q1 fan.

Path dispersion is large *by construction* — the real-GDP ensemble standard deviation is 4.6–5.3 pp, a 5th–95th band spanning ≈ 16 pp — while the Monte-Carlo standard error of the mean is ≈ 0.2 pp: the width is model dispersion, not simulation noise. Per Table 14 the real-GDP 90% band is close to nominal coverage in backtest, so that fan means what it says; the deflator and policy-rate fans are demonstrably too narrow and should be read as lower bounds on uncertainty.

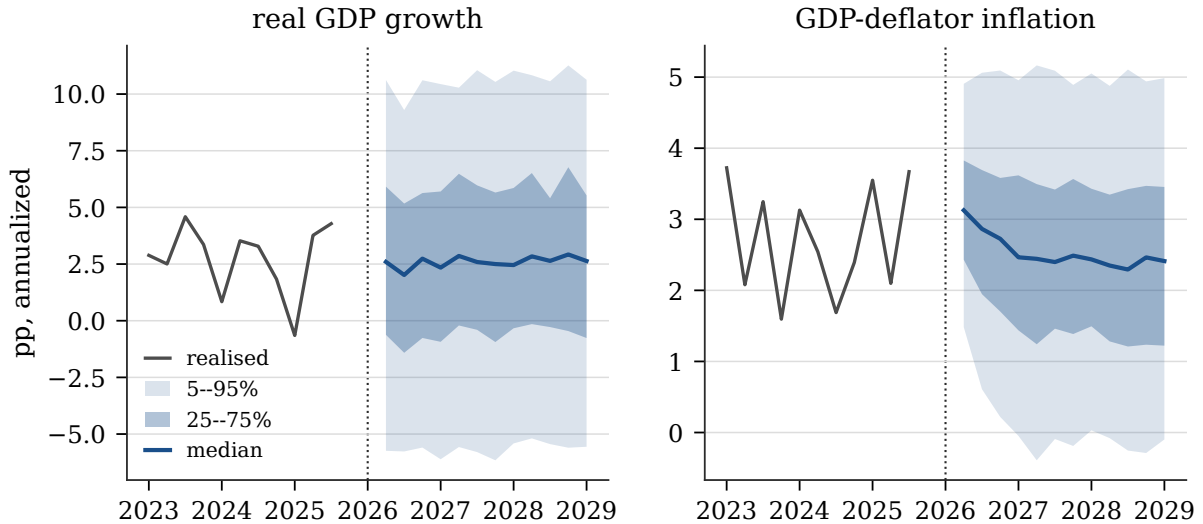


Figure 9: F6 — current outlook from the 2026Q1 origin, unscored. Realised history from 2023Q1 is prepended in grey for context; the fan shows the ensemble median with 25–75th and 5–95th percentile bands over $h = 1 \dots 12$. The dotted line marks the origin. Data: `abm_ensemble_summaries.csv` (outlook run), `quarterly_panel.csv`.

10 Limitations and open problems

Ranked by inferential risk to the results above.

10.1 Revised data, mixed vintage

The entire exercise estimates and scores on a current-vintage panel, and the annual structure is carried backwards to historical origins. The first limitation is symmetric across all twenty columns and matches the reference paper’s own design; the second is asymmetric and ours alone, now bounded for the input–output block by the direct measurement of \$7.8 (moving the structure to 2017 or 2012 changes the headline ratio by at most 0.0007) but still binding for counts, taxes and the opening labour stock. No real-time information set was reconstructed, so nothing here is a claim about what a forecaster would have produced in 2010; the route to that claim is a data-acquisition programme — year-by-year annual history for the 37 structural arrays, plus a real-time truth layer of the Croushore and Stark [2001] kind — not a modelling one.

10.2 The unemployment opening level

Unemployment is scored and lost (§7.6), for a measured reason: every origin opens at the structural year’s labour market. Fetching historical annual Current Population Survey stocks would make the level scoreable at historical origins; the dynamics, repaired by the trend-growth component, are already there. Separately, the labour market lives on a fixed agent pool: with supply growth registered in productivity rather than population, unemployment remains a valid object only within the forecast window — on 24-quarter simulations the pool still exhausts beyond roughly four years. Population growth in the agent count itself is the clean long-run fix and is an intrusive change to the initialisation machinery.

10.3 Interval calibration of the deflator and the policy rate

The one leg of the preregistered bar the model fails. The deflator ensemble is under-dispersed at every level (90 % coverage 0.70–0.79); the policy-rate bands collapse with horizon (0.16 at

$h = 12$). Both are mechanism-level facts — the price block inherits too little of the quantity dispersion, and the Taylor-rule path is too deterministic — and both are named, measured targets for the next model change. Widening by post-processing would be a publication-layer correction, not a model improvement, and is prohibited by the firewall from touching these tables.

10.4 DSGE estimation is posterior-mode, not full Bayes

The reference paper’s DSGE comparison used full Markov-chain Monte Carlo (250 000 draws per origin); ours estimates the mode of the same posterior at every origin and simulates densities at that mode. Stronger than fixed parameters, weaker than full MCMC — parameter uncertainty is absent from the DSGE bands, which flatters their sharpness and is disclosed wherever they are quoted. Full-chain estimation at every origin is a compute programme, not a conceptual gap.

10.5 Scale, and partial density evaluation

All headline results are at scale 10^{-5} (130 firms). An agent-count sweep shows the central results scale-invariant, but 54 of 68 sectors hold exactly one firm, which makes cross-sectional dispersion and tail behaviour unreliable; a production run at $\geq 10^{-4}$ costs roughly $8\times$ the runtime and should precede any density claim finer than the CRPS and coverage reported here. No probability-integral-transform histogram or log score was computed.

10.6 Absent bridges

Three of the eight registered panel targets are unserved: the personal consumption expenditures price index and its core variant need a consumption-basket price bridge the 68-commodity model does not expose, and payroll employment targets an establishment count on a different universe from the model’s firm employment. The model is compared on five of eight targets, and the repository’s semi-structural comparator (whose native set includes the PCE index) cannot be matched cell-for-cell in this field.

10.7 Separation from the statistical family is marginal

Romano–Wolf separates the ABM from both DSGE columns at 5 % in every track, but from the best statistical columns only at $p \approx 0.08$ on weighted losses — pandemic-era origins dominate the loss metric and the historical-mean constant is loss-close there. Margins of a few percent between adjacent statistical columns should not be treated as established; the model confidence set makes the sample’s limited resolving power explicit.

10.8 Reproducibility notes

The pipeline’s legacy baseline builder cannot regenerate its own pre-reconciliation baselines; the checked-in artifacts load and run correctly, and the structural-vintage builder used in §7.8 was golden-tested against the shipped 2024 artifact (bit-exact), so the provenance chain for every scored input is intact. The ensemble caches regenerate byte-for-byte under the current source tree, which was verified for all 3 660 rows of each frozen ablation column before scoring.

11 Conclusion

An agent-based model of the United States, calibrated from public data under a verified no-fitting rule and completed with three structural components — exactly balanced opening accounts,

drift-form growth expectations, and observed trend growth in supply — was evaluated under the reference protocol of Poledna et al. [2023] against a twenty-column field that includes ten statistical benchmarks and two recursively re-estimated DSGE models, with formal inference, exact density scores, matched-seed component attribution, and a structural-vintage robustness sweep. Three findings are worth carrying away.

First, in a structural simulator the accounting is load-bearing, and it is where most of the forecasting accuracy lives. The largest single contributor to the ABM’s standing is not a behavioural rule but an identity: making the opening supply–use system balance exactly, in the model’s own valuation basis, on its own axes. A published table balances only as published; after the valuation bridge and the industry-to-commodity projection a residual survives, and a model whose capacity envelope is anchored to those accounts converts the residual into a permanent, sector-locked rationing regime (−2.75 pp of growth bias, in this calibration). Reduced-form models cannot exhibit this failure class — they have no identities to violate — which cuts both ways: the ABM is exposed to it, and repairing it is a real, attributable source of forecasting skill. The repair also made a falsifiable prediction — a quantity defect should leave the price block untouched — that held to the bit.

Second, a fixed-population, fixed-productivity economy cannot host trend growth, and the stable way to give it one is on the supply side. Without trend supply growth the model must convert demand trend into employment trend until the labour pool is exhausted, and output trend into utilisation until capacity binds — both measured, both terminal. Registering observed productivity-plus-labour-force growth in the supply coefficients repairs the labour market’s dynamics and lifts the capacity ceiling while leaving headline accuracy essentially unchanged — evidence that the model’s forecasting skill was never the artifact of a supply pathology. The demand-side alternative, investment responding to the capacity gap, destabilises the economy in two independent designs: a rationed goods market with adaptive expectations amplifies any injected investment demand into a boom. That fragility is a genuine property of the model class at this calibration, and it marks stable endogenous investment-driven growth as the design problem standing between this model and multi-year use.

Third, against the strongest available challengers the ABM’s advantage is real, specific, and honestly bounded. The margin over a fully re-estimated Smets–Wouters system — itself granted more data than any other column, and better than every statistical model — is significant under multiplicity control in every track, and it is concentrated where an agent-based model ought to earn it: real activity at business-cycle horizons, with inflation accuracy statistically indistinguishable and the density advantage carried by sharpness. Against simple statistical models the separation is marginal on this sample. The deflator’s intervals are too narrow, and that single calibration leg fails the preregistered superiority bar, so the claim published is the conditional one of §7.1 — which is also, precisely, what the evidence supports. What stands between this diagnostic and a real-time forecasting claim is not further tuning — the firewall forbids it — but data: vintage-correct structural inputs, origin-correct opening labour stocks, and full-posterior DSGE baselines. Each is named, none is done, and the apparatus to score them already exists.

A Reproducibility

A.1 Commands

All commands run from the repository root. Both environment variables are hard requirements on the benchmark path: the runner raises an error unless the Julia and BLAS thread counts are both 1.

```
# 1. build the reconciled calibration artifact (~40s)
julia -project=scripts/us \
```



```

scripts/us/calibration/reconcile_commodity_balance.jl \
-mode=final_demand_scaled -expectations=rw_drift
# 2. the ABM ensembles: the model and its ablation columns,
# 61 origins x 500 paths each
JULIA_NUM_THREADS=1 OPENBLAS_NUM_THREADS=1 julia -project=scripts/us \
scripts/us/forecasting/diagnostics/abm_revised_comparison/\
run_revised_data_abm_comparison.jl \
output/us_forecasting/stage2b_full/v3_headline 500 headline_v3
... v2_headline 500 headline_v2 ... v1_headline 500 headline
... v3_ar1 500 headline_v3_ar1
... outlook_v3 500 outlook_v3

# 3. the DSGE columns: per-origin posterior-mode estimation + 500-path densities
JULIA_NUM_THREADS=1 OPENBLAS_NUM_THREADS=1 julia -startup-file=no \
-project=scripts/us scripts/us/forecasting/benchmarks/dsge_columns/\
run_dsge_columns.jl \
output/us_forecasting/stage2b_dsge_columns both

# 4. the composite twenty-column scorecard, densities included
.../stage2b/run_stage2b_scorecard.jl \
output/us_forecasting/stage2b_scorecard \
-v1=.../v1_headline -v2=.../v2_headline -v3=.../v3_headline \
-dsge=.../stage2b_dsge_columns

# 5. inference (HLN-DM, Romano-Wolf, MCS), verdict, and rendered tables
.../stage2b/run_stage2b_inference.jl output/us_forecasting/stage2b_scorecard
.../stage2b/verdict_stage2b.jl output/us_forecasting/stage2b_scorecard
.../stage2b/render_stage2b_tables.jl output/us_forecasting/stage2b_scorecard

# 6. structural-vintage robustness (2017 / 2012 artifacts, then re-run step 2 with
-calibration=data/us/calibration/US_2017_calibration_object_reconciled.jld2)

# 7. figures and this paper
python3 paper/figures_src/make_figures.py
cd paper && latexmk -pdf US_ABM_Forecasting_Paper.tex

```

Step 2 resumes from its own ensemble cache, so an interrupted run continues rather than restarting; the composite scorecard of step 4 is a re-score off the caches (seconds), and it asserts full origin coverage for every column before scoring. Every figure is regenerated by step 7 directly from committed CSVs; no figure contains a hand-entered number, and the generator pins `SOURCE_DATE_EPOCH` so a re-run reproduces each figure byte for byte.

A.2 Artifact paths

A.3 Identity seals

Every run manifest seals the identities that could move a number: the calibration artifact path and SHA-256 (main artifact `data/us/calibration/US_2024_calibration_object_reconciled.jld2`, digest `57e23f4a...e514760`), the reconciliation mode and λ (`final_demand_scaled`, `0.9834601934561227`), the expectation specification (`rw_drift`), the panel SHA-256 (`f7bb26a4...bff851bbe`), the sealed numerical kernel and source-tree digests, path counts (500 / burn-in 0 / 12 simulated quarters), and the Julia version and thread counts (1.10.3 / 1 / 1). The trend-growth fixture and the DSGE column provenance files carry their own SHA-256 sums and retrieval timestamps. The λ semantics are recorded verbatim in each manifest: *“explicit accounting choice: the four final-demand aggregates C , G , I and X (and capital consumption and gross capital formation dwellings, which set the investment budget) are scaled by λ so the artifact’s expenditure aggregates match its production account and the opening commodity balance clears exactly. λ is fixed by the accounting identity alone and was not chosen with reference to any forecast error.”*

Path (under the repository root)	Contents
output/us_forecasting/stage2b_scorecard/	the twenty-column scorecard: weighted/relative/summary point tables, per-cell losses, CRPS, coverage, unemployment dispersion, HLN-DM, Romano–Wolf, MCS, the verdict, and STAGE2B_TABLES.md
output/us_forecasting/stage2b_full/	ensemble caches and per-run scorecards: v3_headline (the model), v2_headline / v1_headline / v3_ar1 (ablation columns), v3_struct2017 / v3_struct2012 (vintage robustness), outlook_v3, each with retained predictive draws
output/us_forecasting/stage2b_dsge_columns/	per-origin DSGE posterior modes, estimation status, ensemble summaries and retained draws for both columns
output/us_forecasting/commodity_balance_reconciliation/	reconciliation_report_rho1.txt,
output/us_forecasting/commodity_balance_reconciliation_structural_vintage/	reconciliation_by_commodity_rho1.csv
data/us/calibration/US_2024_calibration_object_reconciled.jld2	reconciliation reports for the 2017 and 2012 structures
scripts/us/forecasting/diagnostics/stage2b/STAGE2B_PROTOCOL.md	the calibration artifact (with _2017_/_2012_ variants and provenance files beside each)
scripts/us/forecasting/benchmarks/dsge_columns/DSGE_COLUMNS_MODEL_CARD.md	the frozen protocol: scored columns, components, calibration rules, success bar
scripts/us/forecasting/diagnostics/abm_revised_comparison/trend_growth_series.csv	DSGE column conventions, estimation settings and disclosures
scripts/us/forecasting/diagnostics/revised_data/fixtures/quarterly_panel.csv	the trend-growth calibration series with SHA-pinned provenance
	the scoring panel: 101 quarters, 2000Q3–2025Q3

A.4 Seed policy

- Simulation seeds are $\text{hash}((:\text{abm_revised_v1}, \text{origin}, \text{path})) \bmod 2^{30}$: domain-separated by origin and path, so no two cells share a stream.
- Every ablation and robustness variant draws the *identical* seed stream as the model, which makes each contrast matched-seed; the expectation branch consumes exactly one Normal variate either way to preserve the alignment.
- The seed is set *immediately before* model construction, because the initial microstate is drawn from the global generator; seeding after construction yields a random initial state and is a silent error. A fresh model is constructed per path.
- The DSGE columns use fixed per-origin seed bases ($7\,000\,000 + \text{origin index}$ and $8\,000\,000 + \text{origin index}$) with an SHA-domain-separated per-path generator, so their draws are version-independent and reproducible.
- Failed paths are counted and left visible, never resampled. The count is 0 in every run.

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